

Cross-Wearer Gesture Recognition — LOUO Variant Comparison

Generated 2026-05-14 20:23:19

Data & configuration

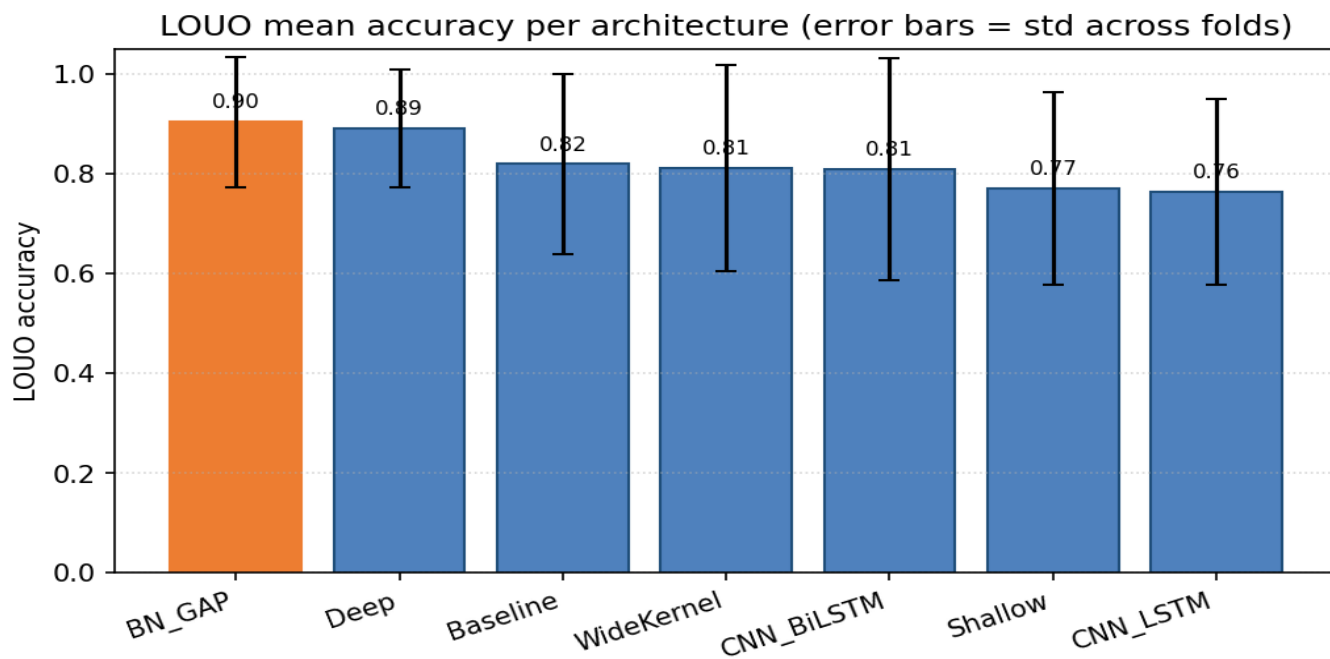
LOUO path 1	/home/jestin/ThesisRepo/ML/NewTestData/1_Alan/Dynamic
LOUO path 2	/home/jestin/ThesisRepo/ML/NewTestData/2_Alex/Dynamic
LOUO path 3	/home/jestin/ThesisRepo/ML/NewTestData/3_Anghad/Dynamic
LOUO path 4	/home/jestin/ThesisRepo/ML/NewTestData/4_Daniel/Dynamic
LOUO path 5	/home/jestin/ThesisRepo/ML/NewTestData/6_Harry/Dynamic
LOUO path 6	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Dynamic
LOUO path 7	/home/jestin/ThesisRepo/ML/NewTestData/11_Mansh/Dynamic
LOUO path 8	/home/jestin/ThesisRepo/ML/NewTestData/12_Marcus/Dynamic
LOUO path 9	/home/jestin/ThesisRepo/ML/NewTestData/14_StephenV2/Dynamic
LOUO path 10	/home/jestin/ThesisRepo/ML/NewTestData/15_Tash/Dynamic
LOUO path 11	/home/jestin/ThesisRepo/ML/NewTestData/16_Bella/Dynamic
LOUO path 12	/home/jestin/ThesisRepo/ML/NewTestData/17_Raquel/Dynamic
LOUO path 13	/home/jestin/ThesisRepo/ML/NewTestData/18_Bridgette/Dynamic
LOUO users loaded	13 (11_Mansh, 12_Marcus, 14_StephenV2, 15_Tash, 16_Bella, 17_Raquel, 18_Bridgette, 1_Alan, 2_Alex, 3_Anghad, 4_Daniel, 6_Harry, 8_Jestin)
Excluded classes	Double_Nothing
Classes	6: Double_Lower, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll
Sensor channels	176 (sequence length 90)
Resample / filter	90 steps · Butterworth cutoff 10.0 Hz, order 4
Per-trial normalisation	none
Downstream normalisation	minmax
Sampling rate	30.0 Hz
Random state	42
Augmentation	on · copies=2 · amplitude_scale, baseline_offset
Epochs · batch size	40 · 16
Early stopping	on · monitor val_loss · patience=8
LOUO inner-validation	user
LOUO variants	Baseline, Shallow, Deep, BN_GAP, WideKernel, CNN_LSTM, CNN_BiLSTM
Final-retrain variant	BN_GAP_Wide
Final-retrain held-out user	— (trained on all users)

Variant comparison (LOUO)

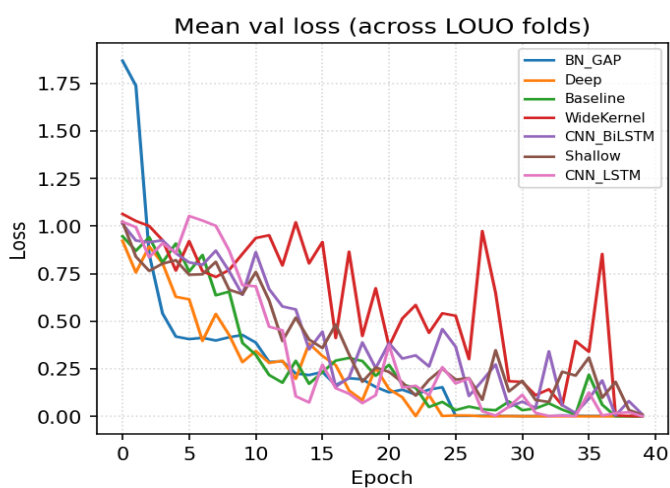
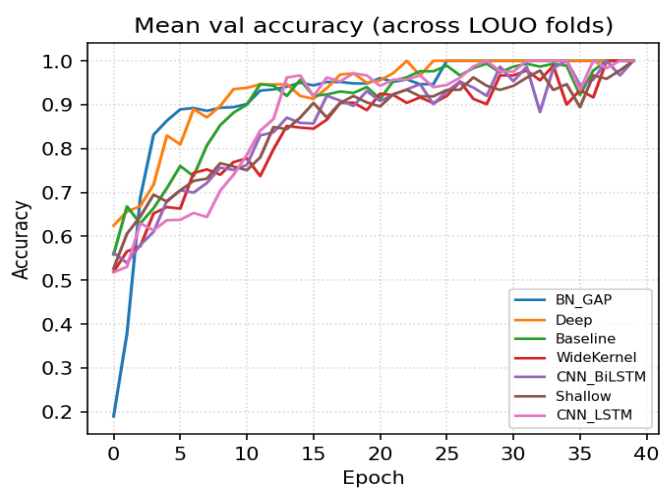
Variant	Params	LOUO mean	LOUO std	Folds	Held-out user acc
BN_GAP	57,190	0.9033	0.1309	13	—
Deep	203,878	0.8904	0.1178	13	—
Baseline	113,190	0.8186	0.1802	13	—
WideKernel	131,622	0.8109	0.2070	13	—
CNN_BiLSTM	105,510	0.8083	0.2229	13	—
Shallow	72,454	0.7692	0.1928	13	—
CNN_LSTM	68,390	0.7628	0.1861	13	—

Best LOUO mean: BN_GAP

LOUO accuracy per variant

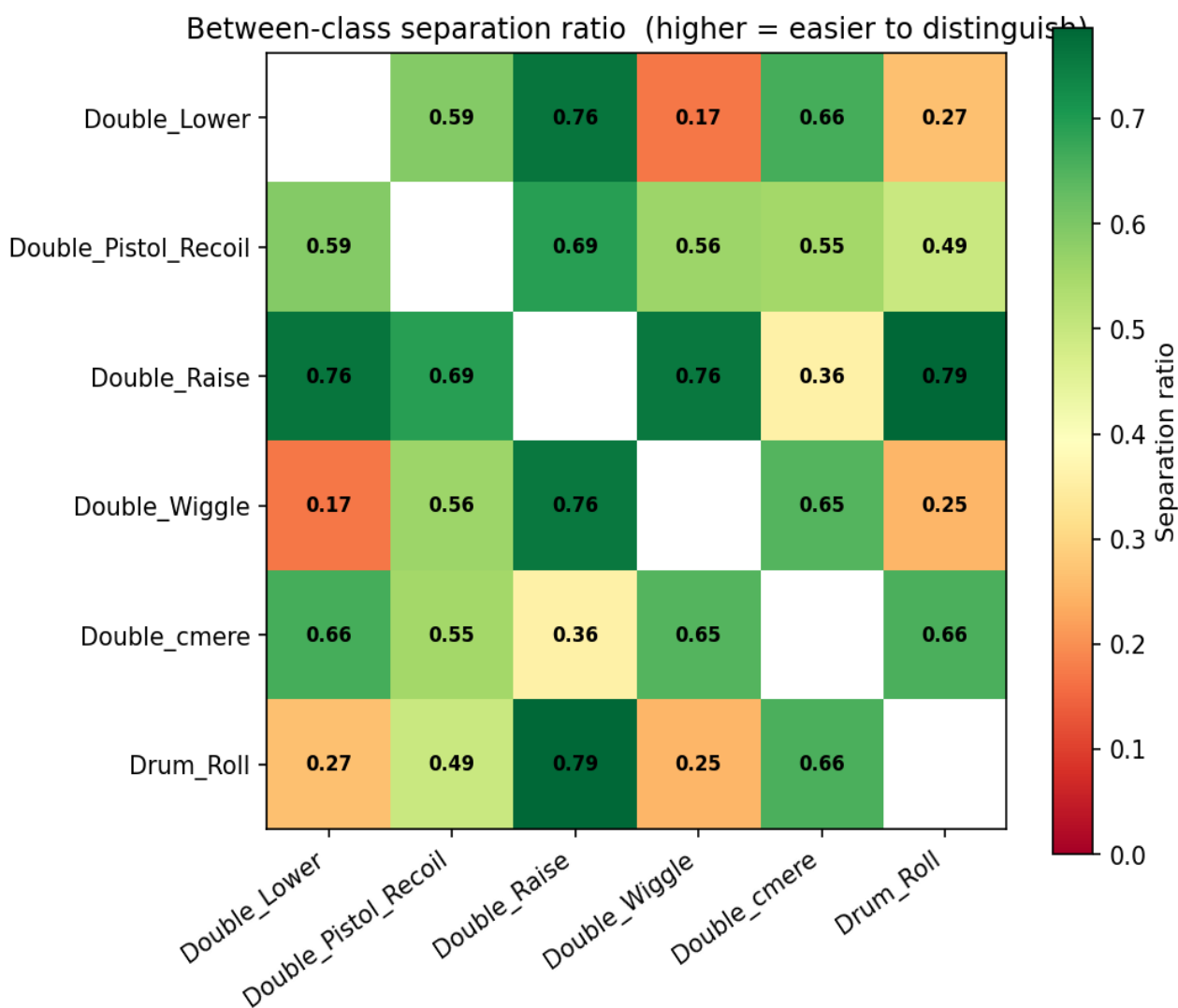
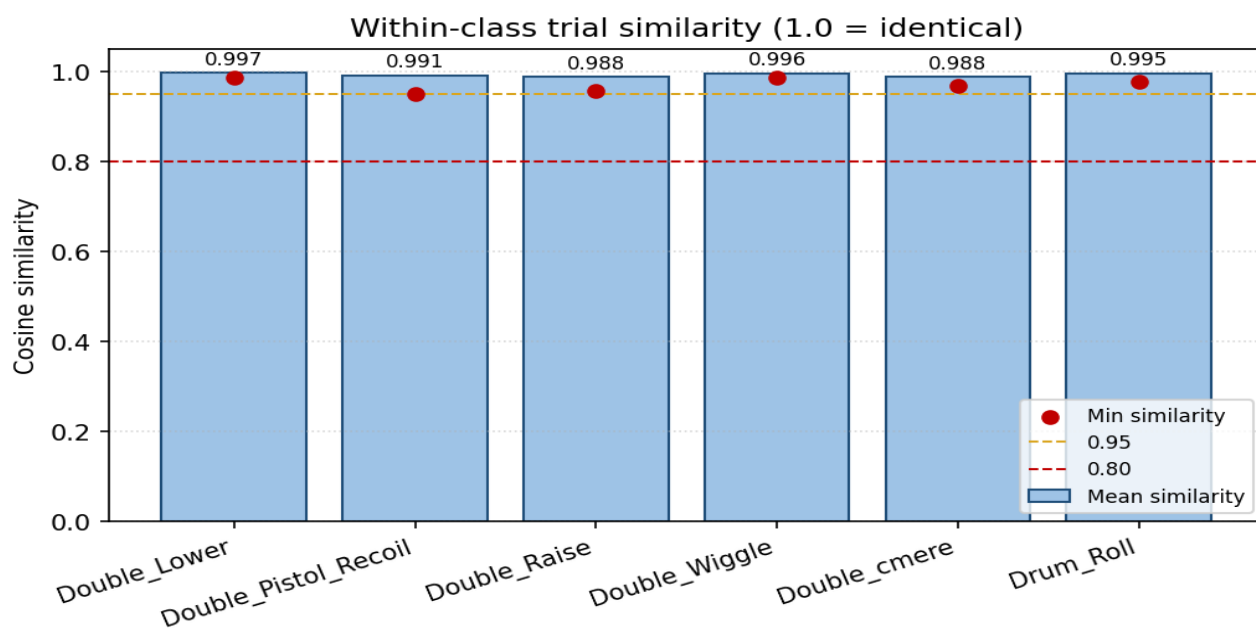


Mean training curves across LOUO folds

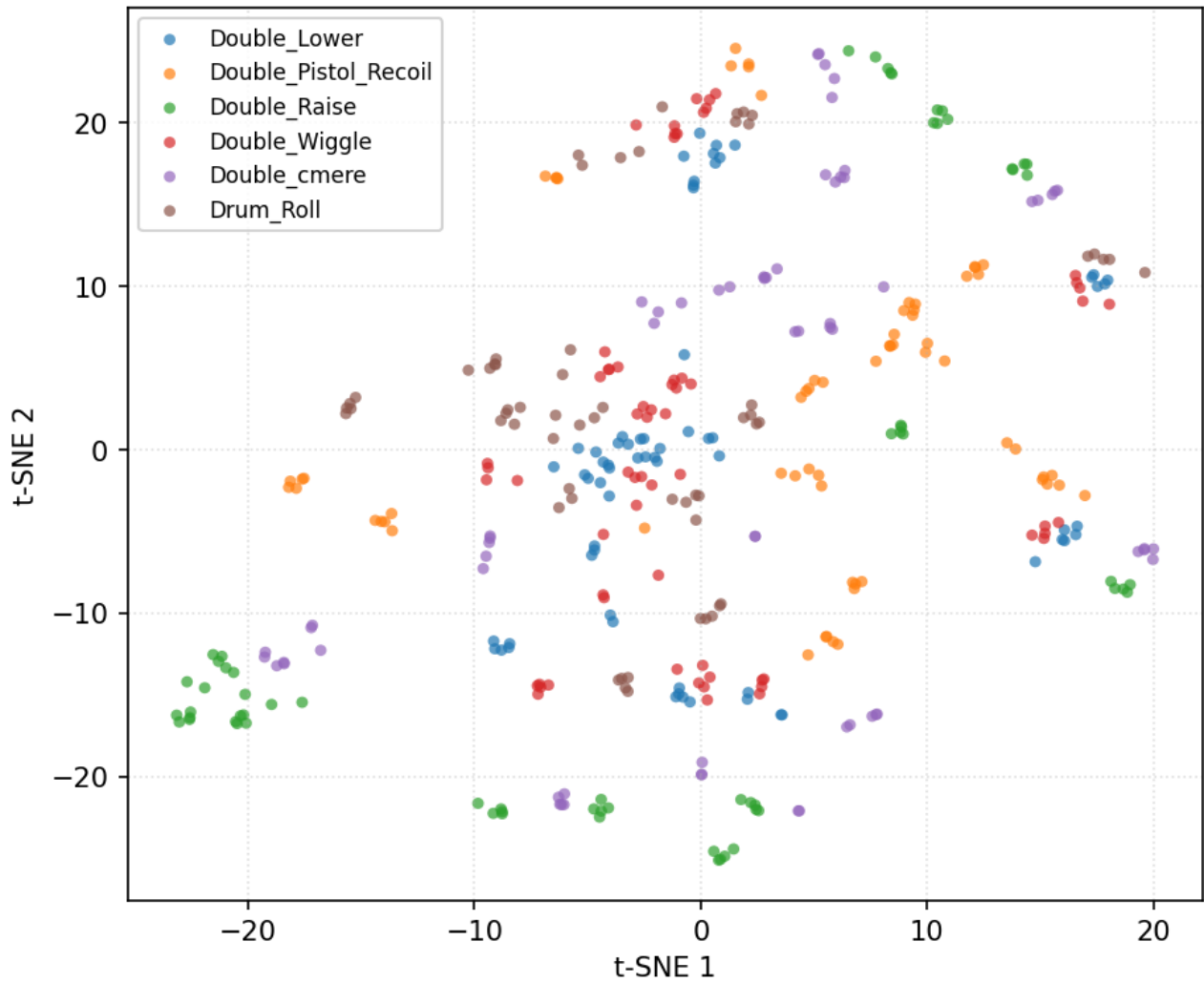


Data diagnostics — pooled across users

Filtered + resampled trials from every LOUO user (before per-trial normalisation).



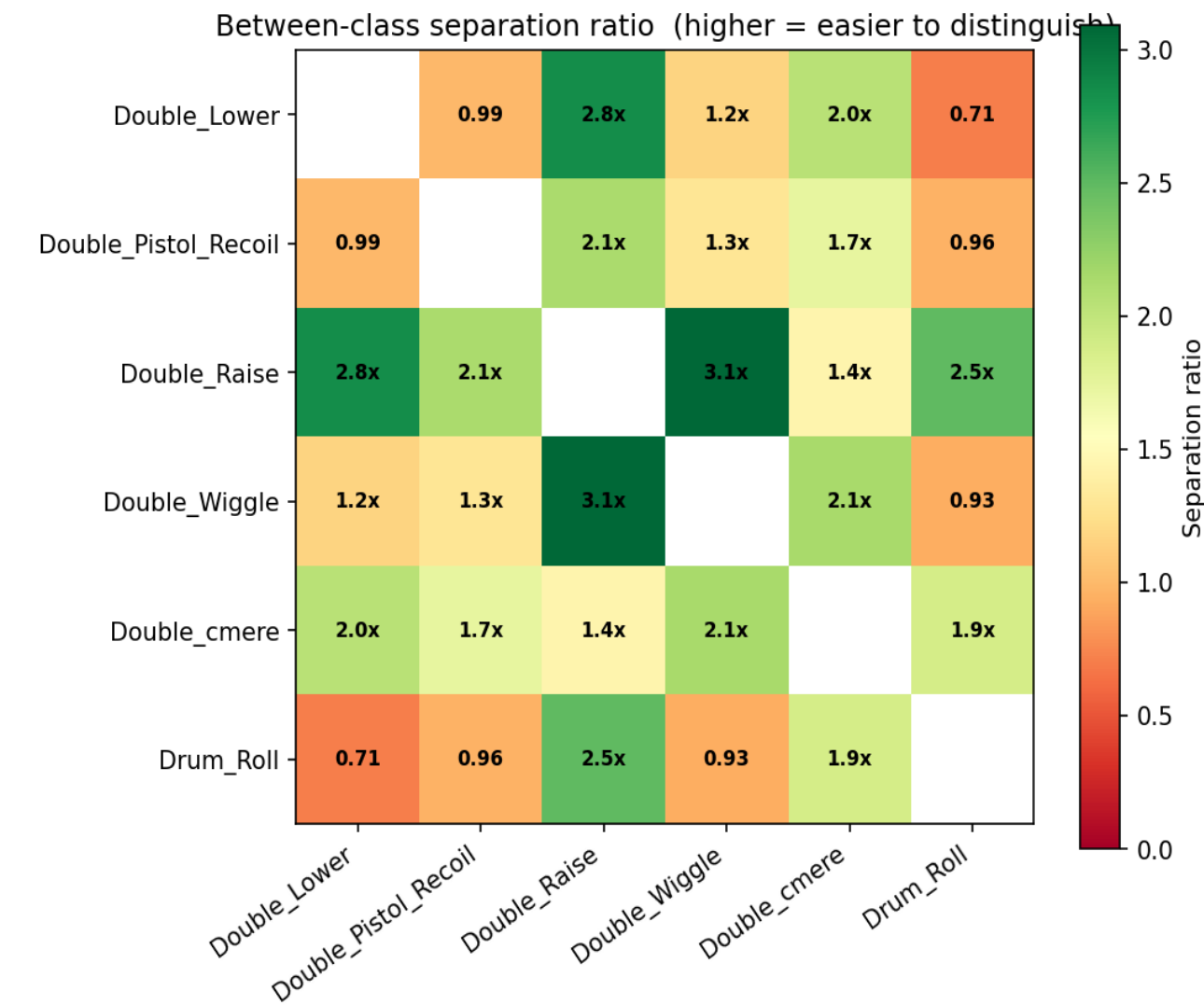
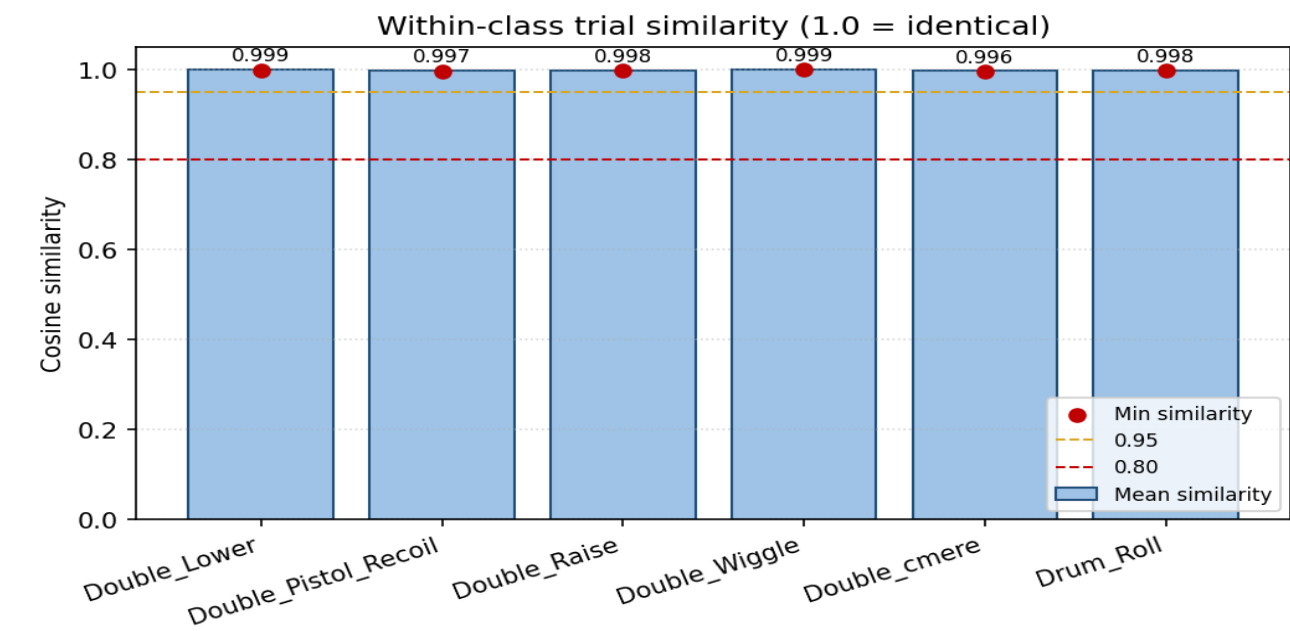
t-SNE feature space projection



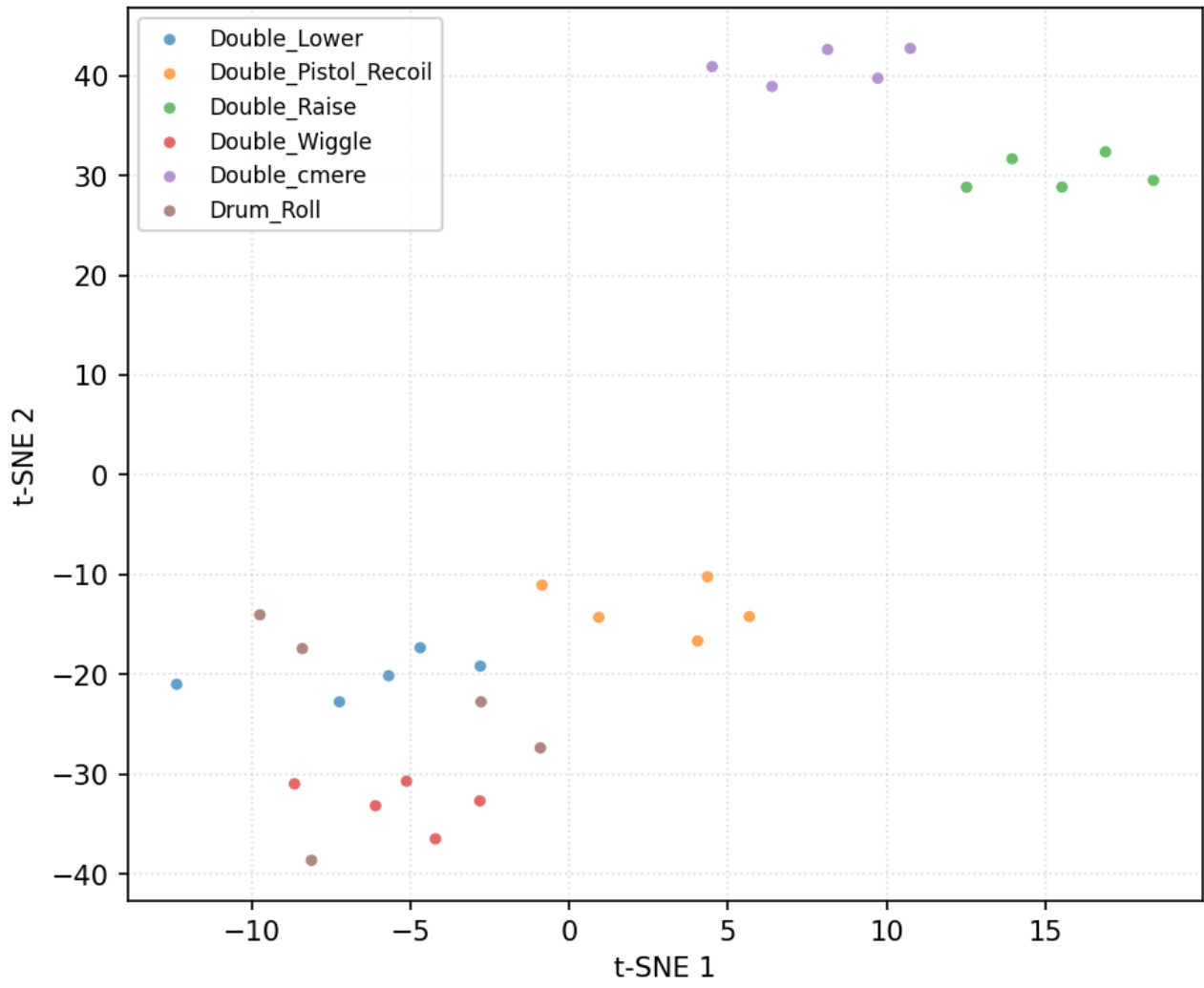
Reading guide: within-class similarity > 0.95 = very consistent; < 0.80 = high variance. Separation ratio > 100x = trivially separable, 20-100x = easy, 5-20x = moderate, < 5x = hard (expect confusion).

Data diagnostics — user 11_Mansh

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

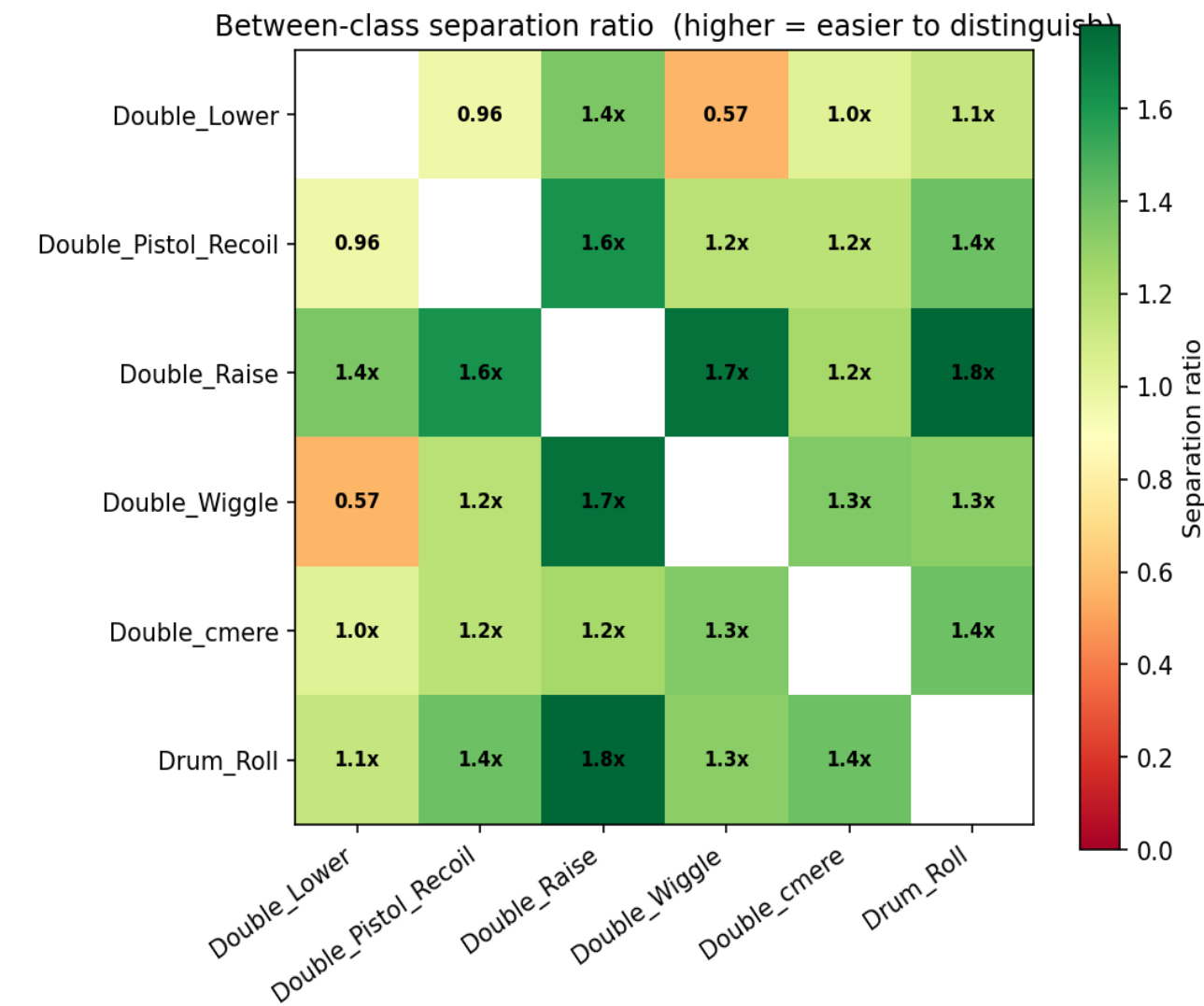
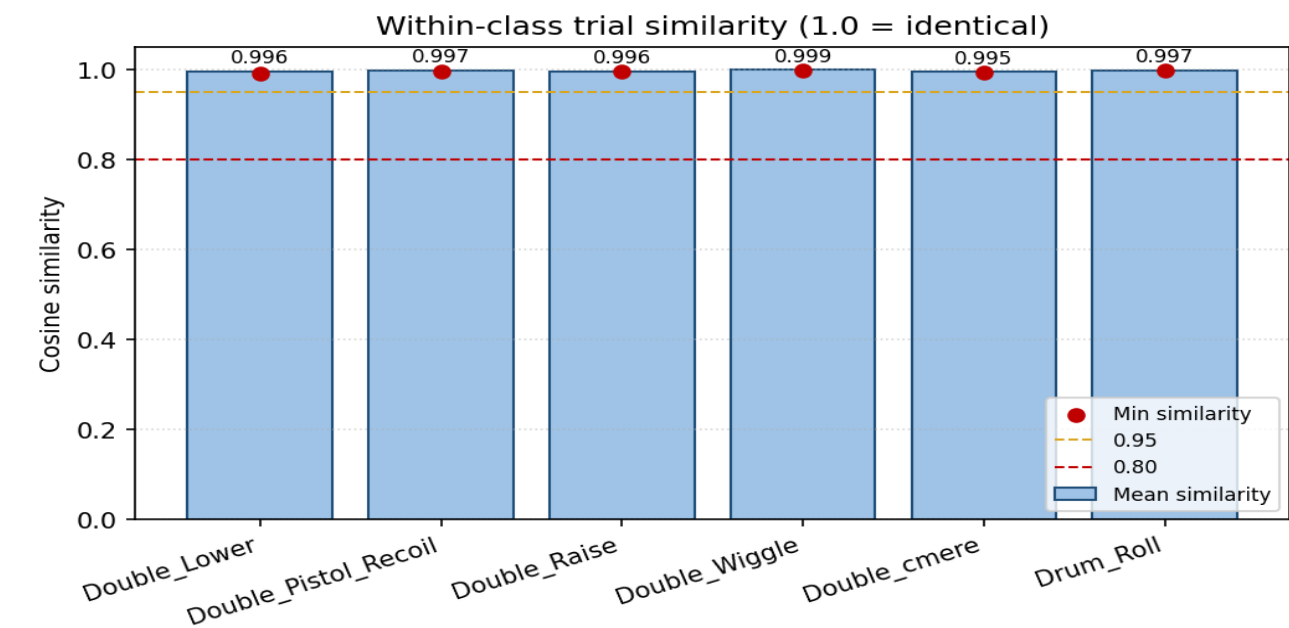


t-SNE feature space projection

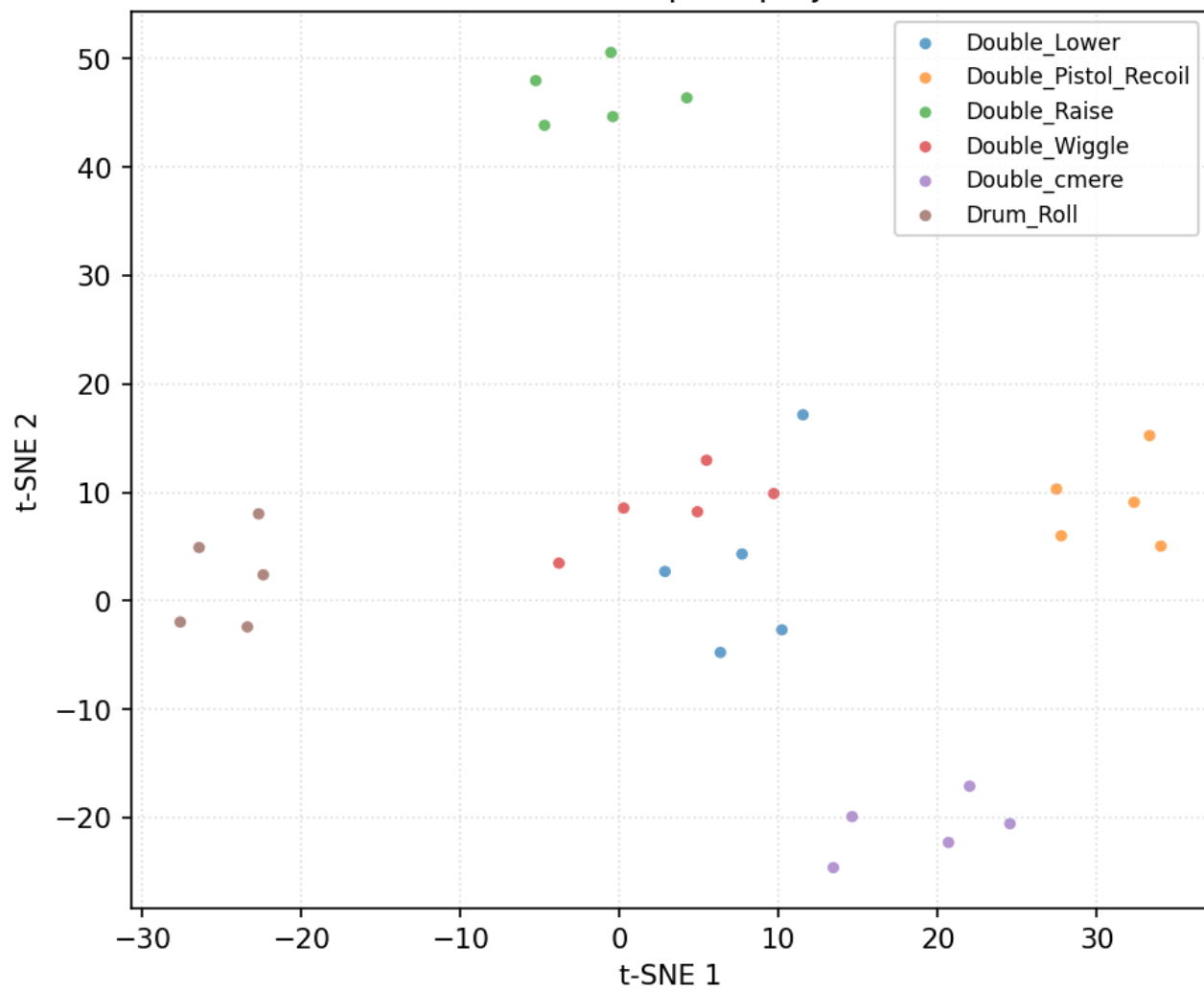


Data diagnostics — user 12_Marcus

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

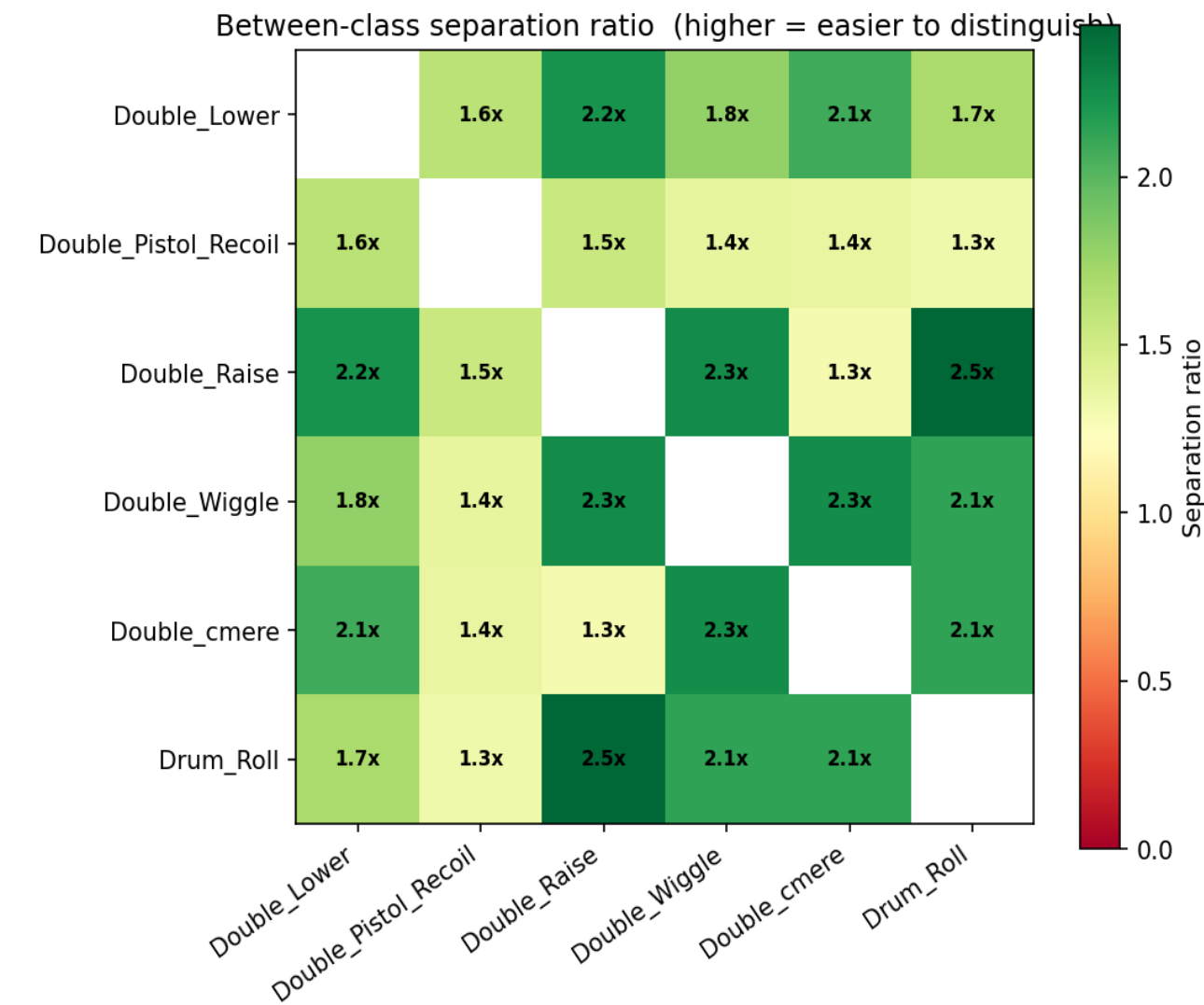
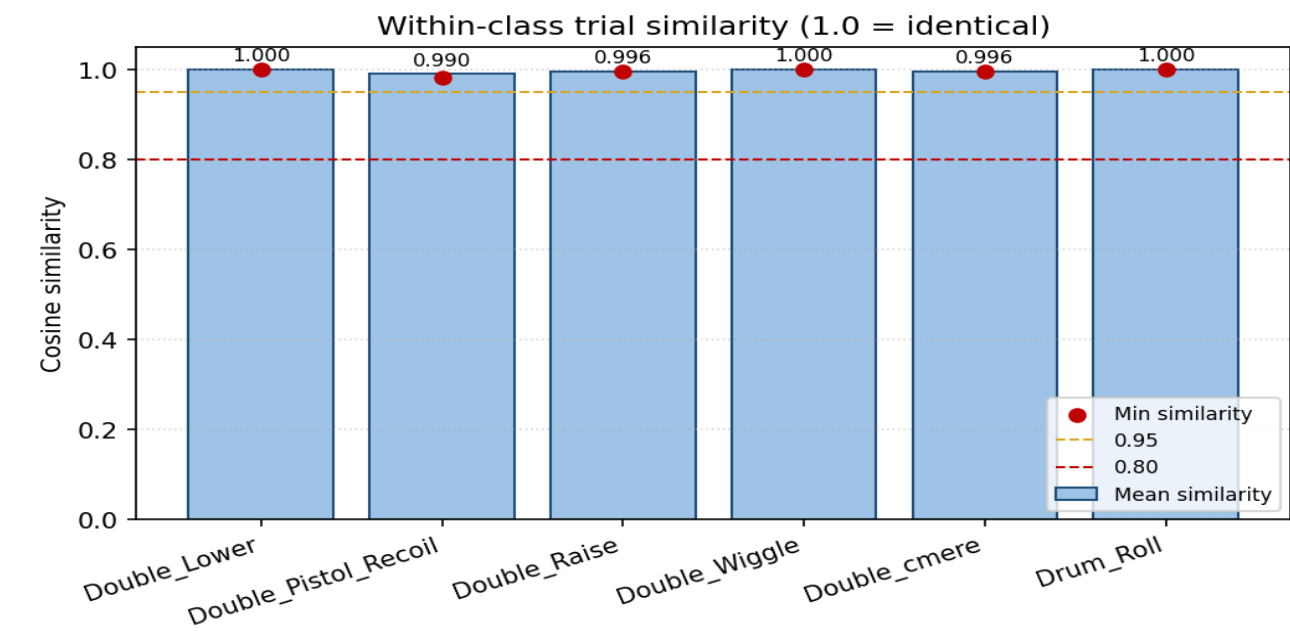


t-SNE feature space projection

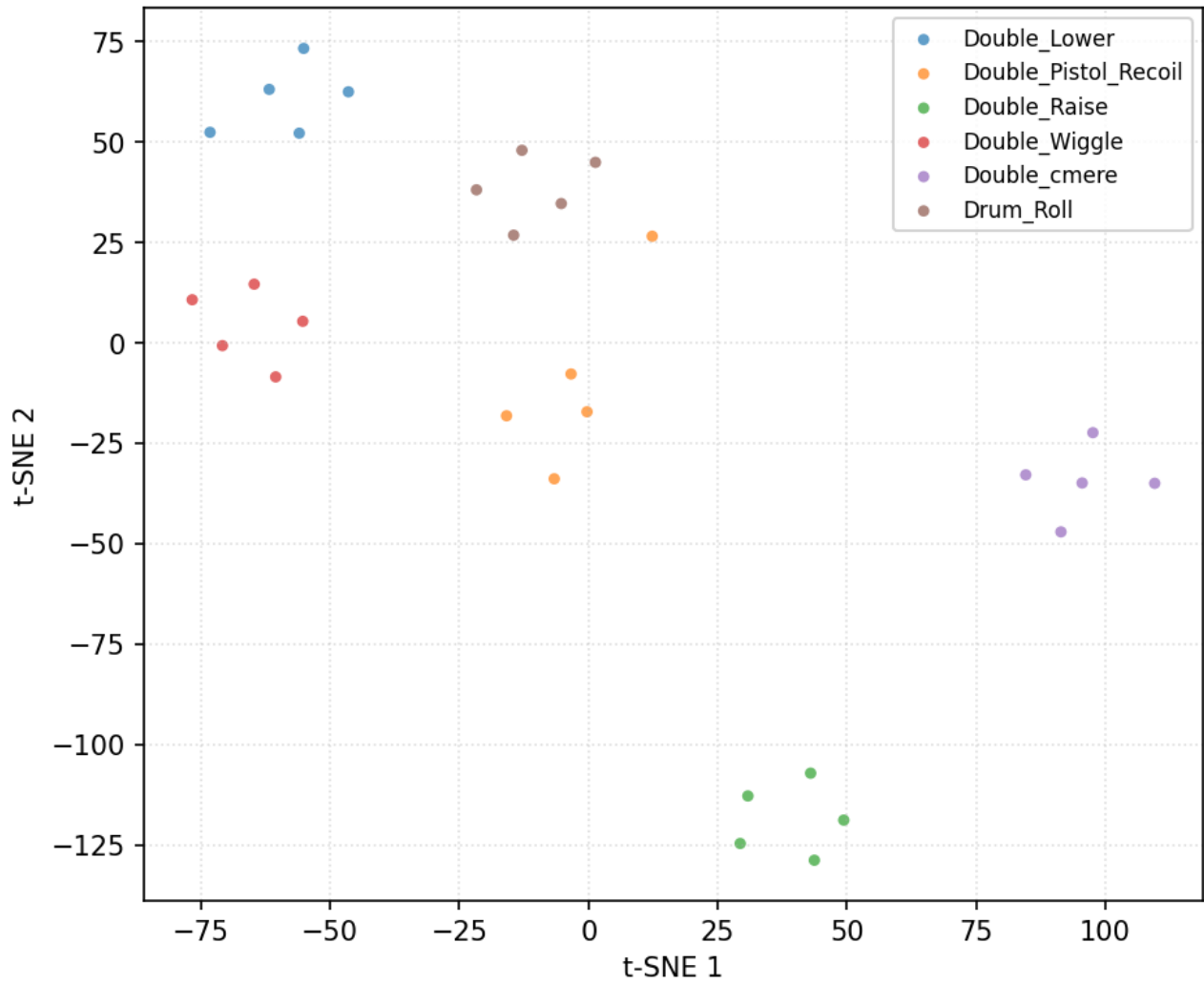


Data diagnostics — user 14_StephenV2

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

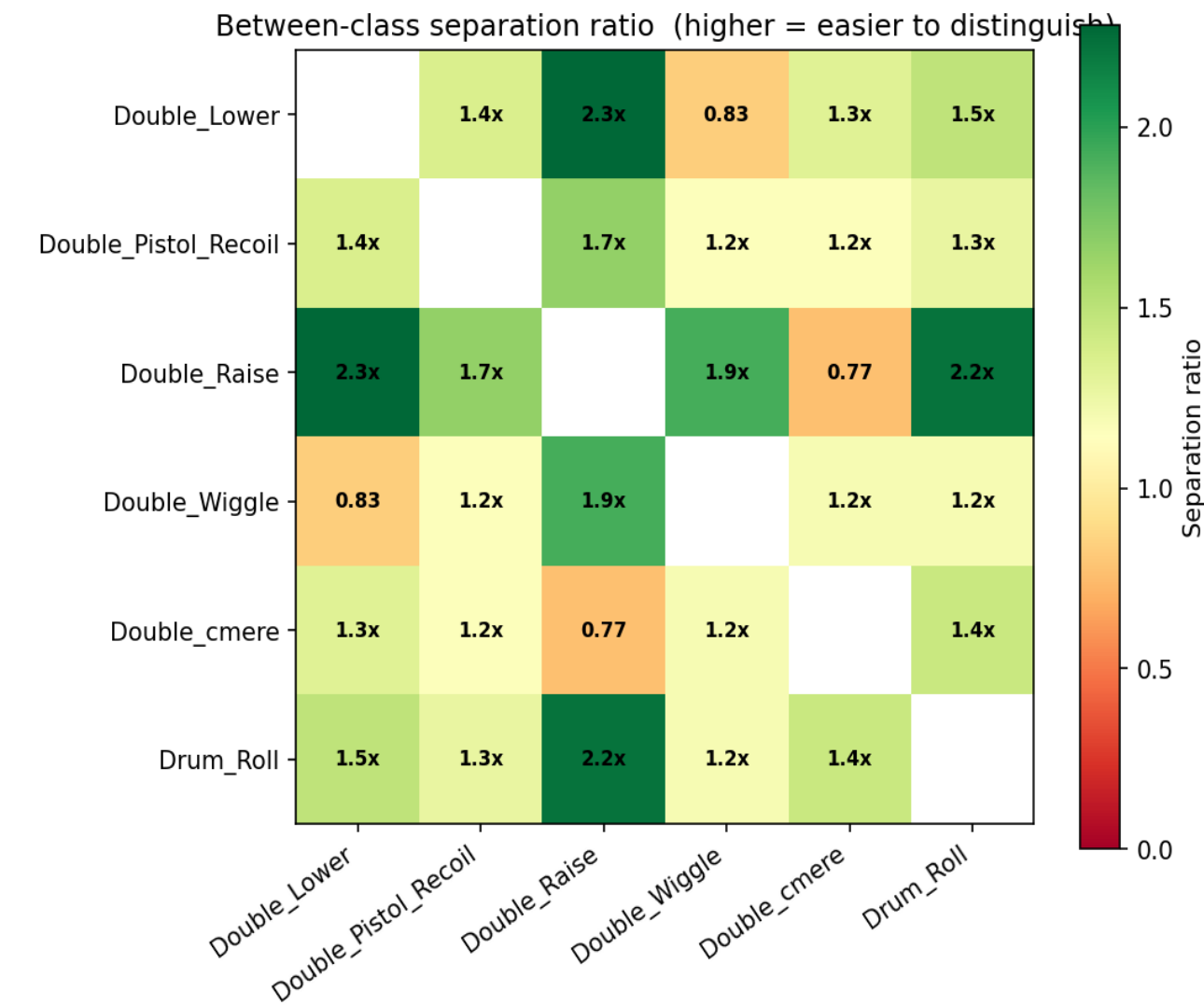
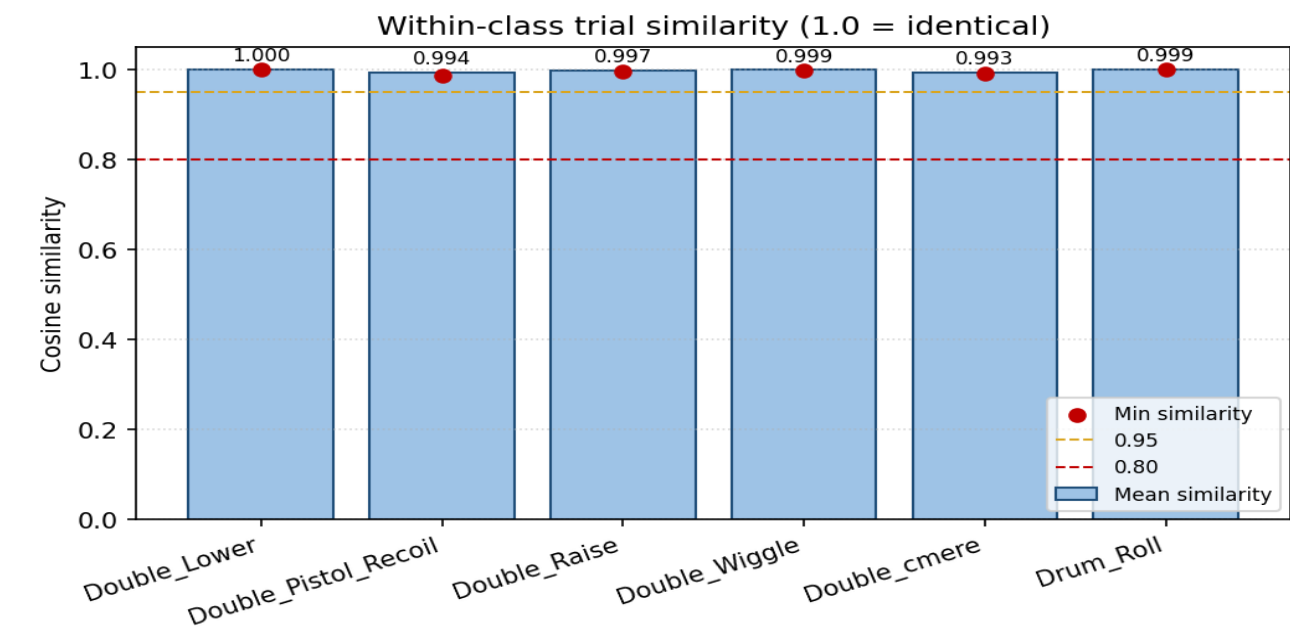


t-SNE feature space projection

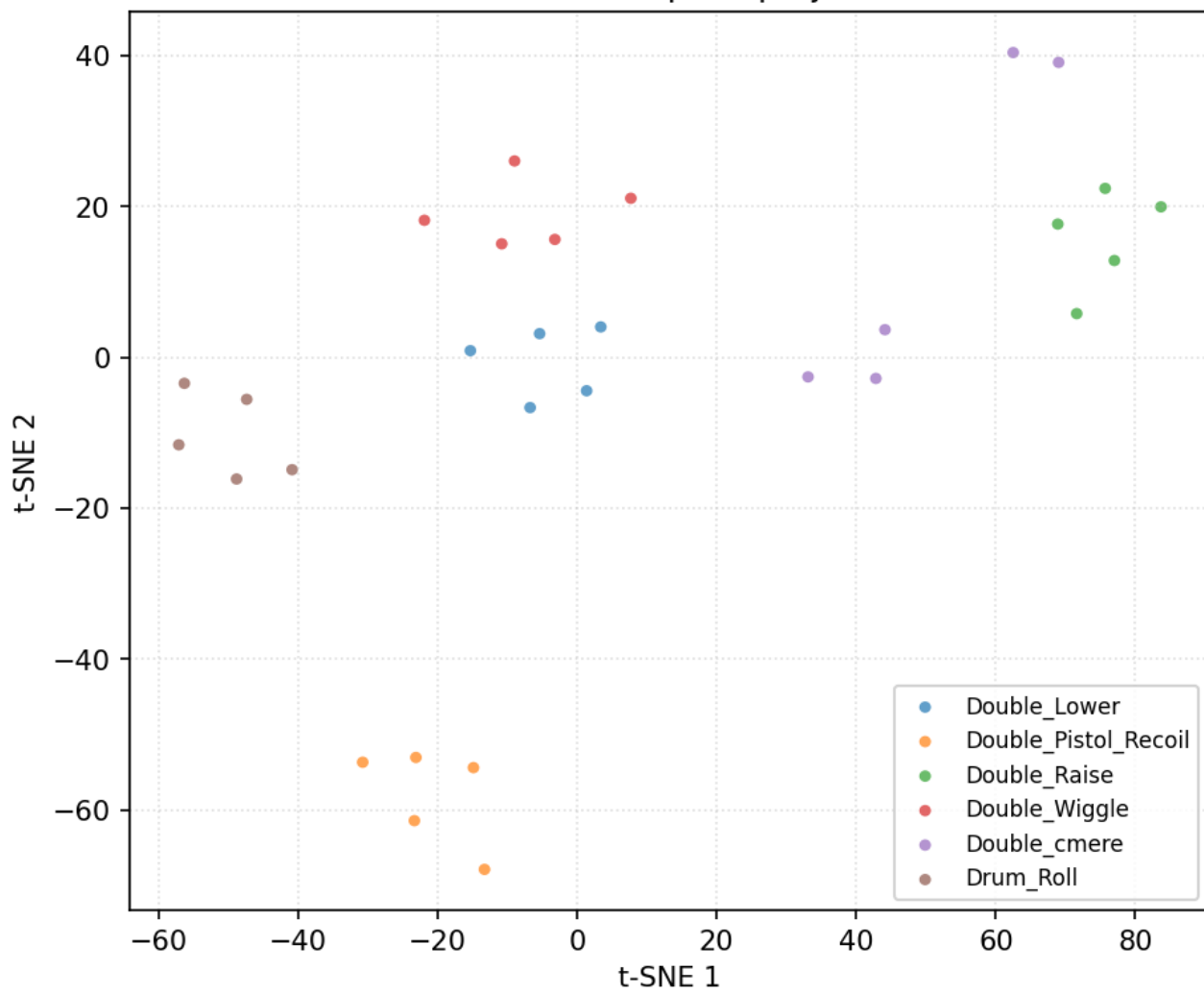


Data diagnostics — user 15_Tash

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

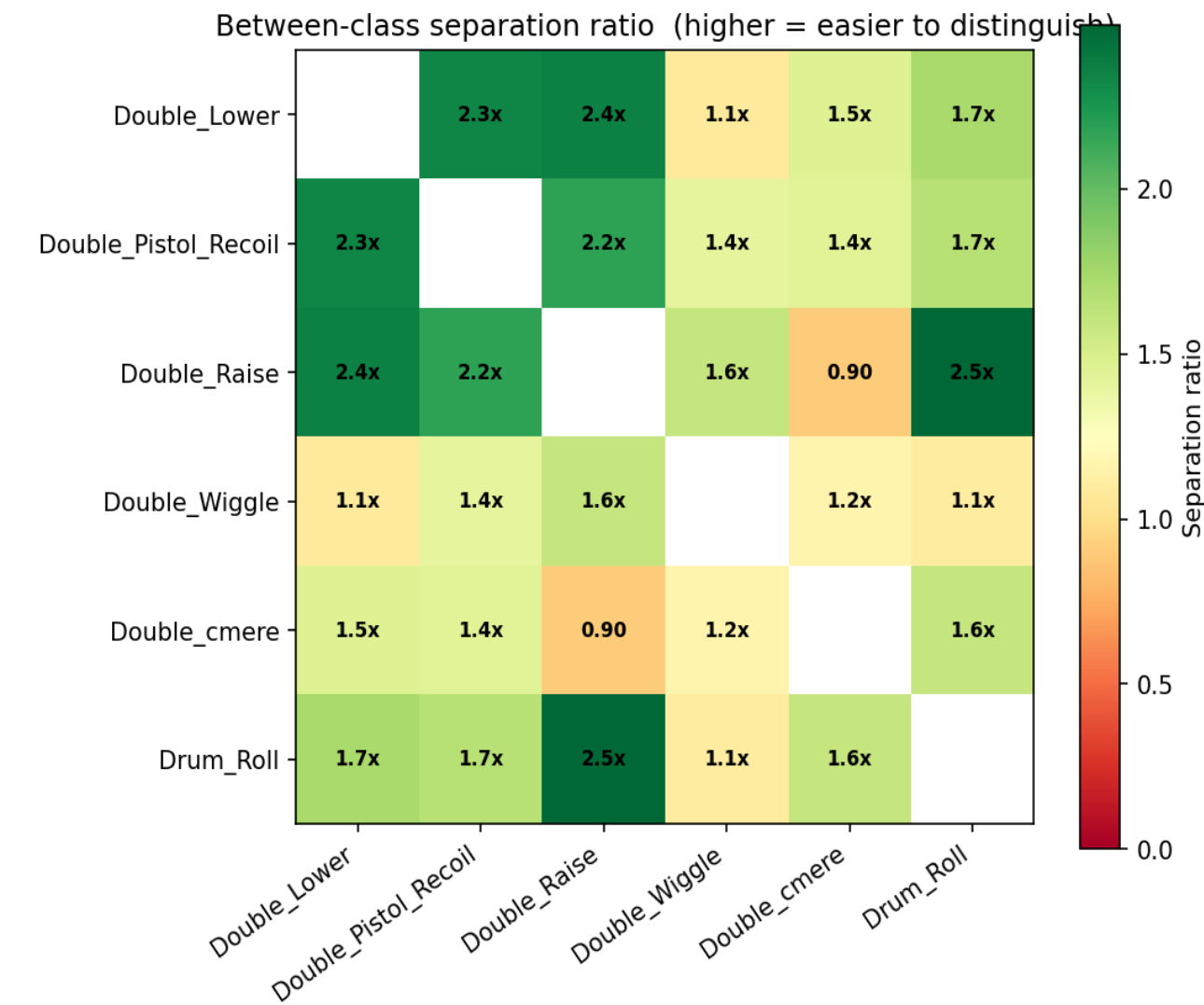
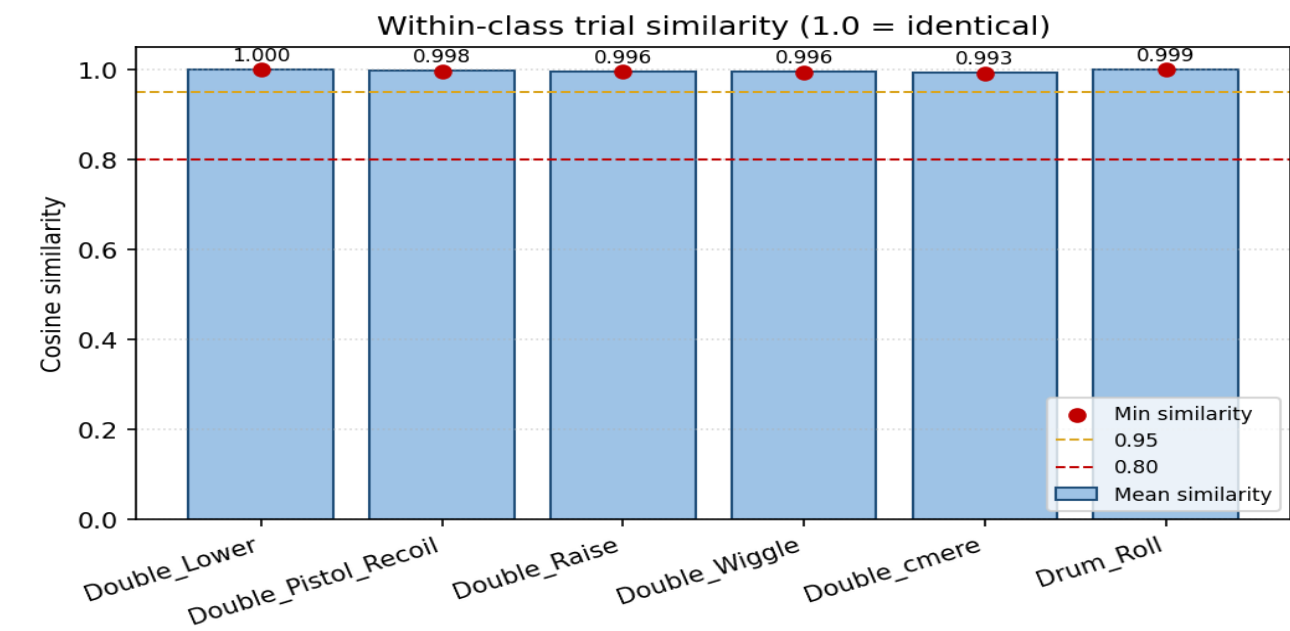


t-SNE feature space projection

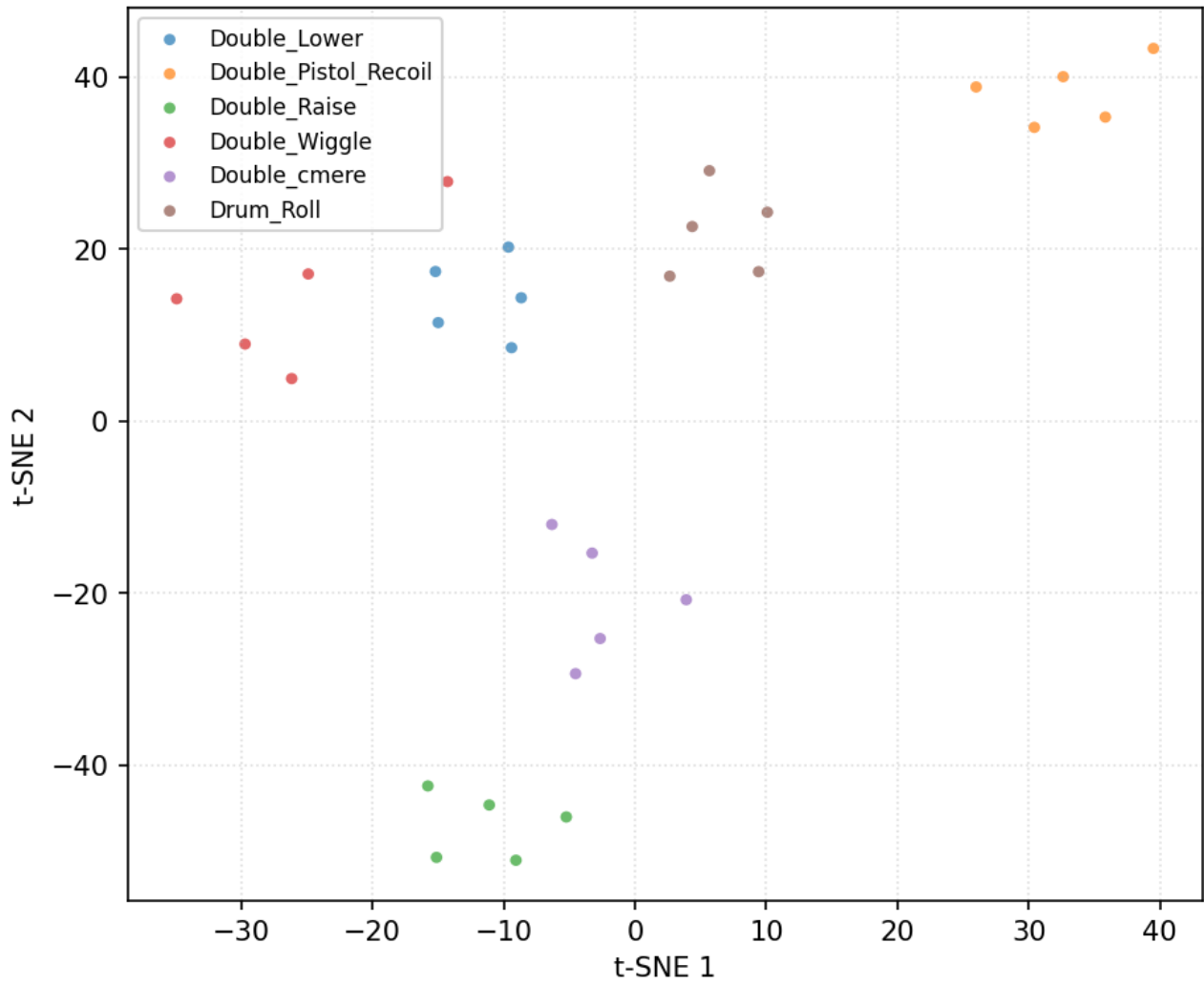


Data diagnostics — user 16_Bella

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

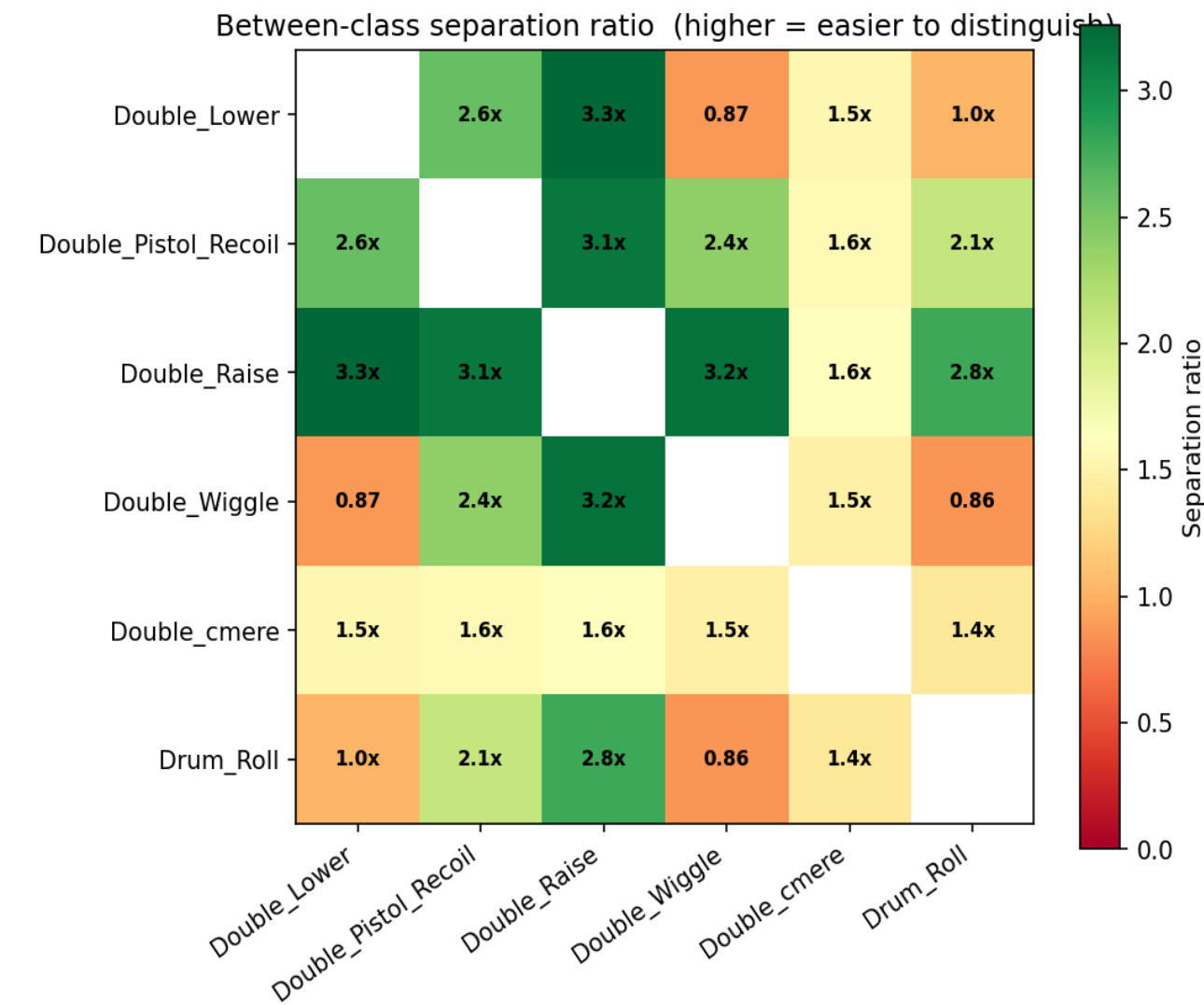
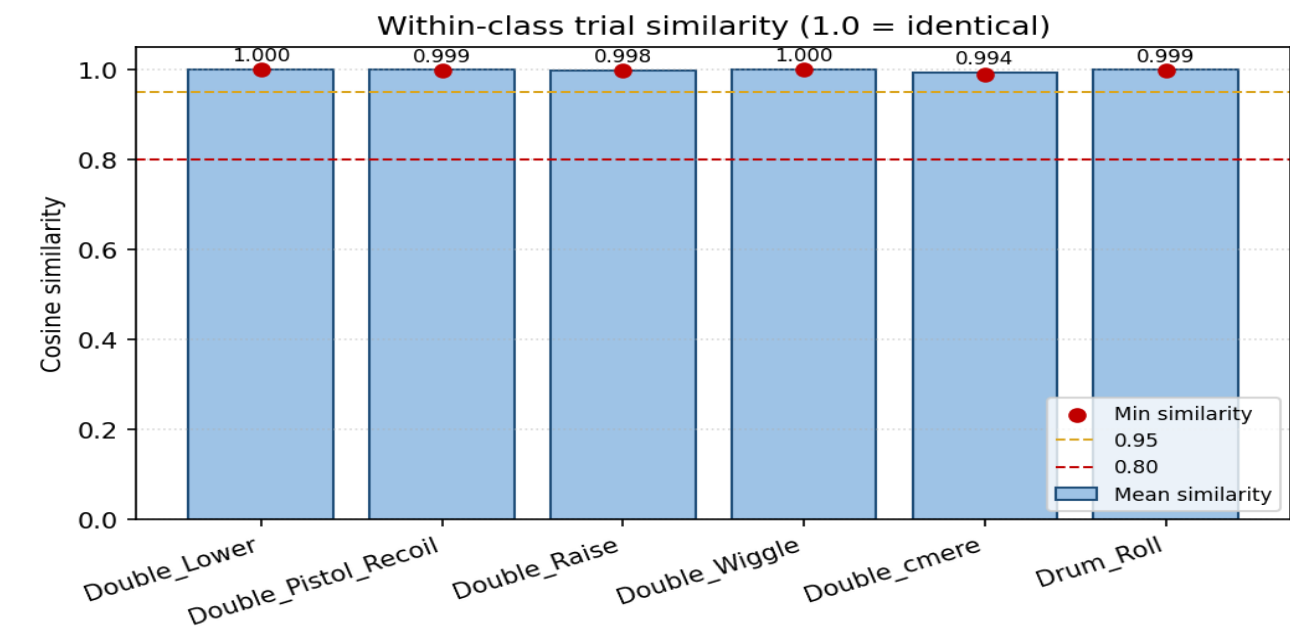


t-SNE feature space projection

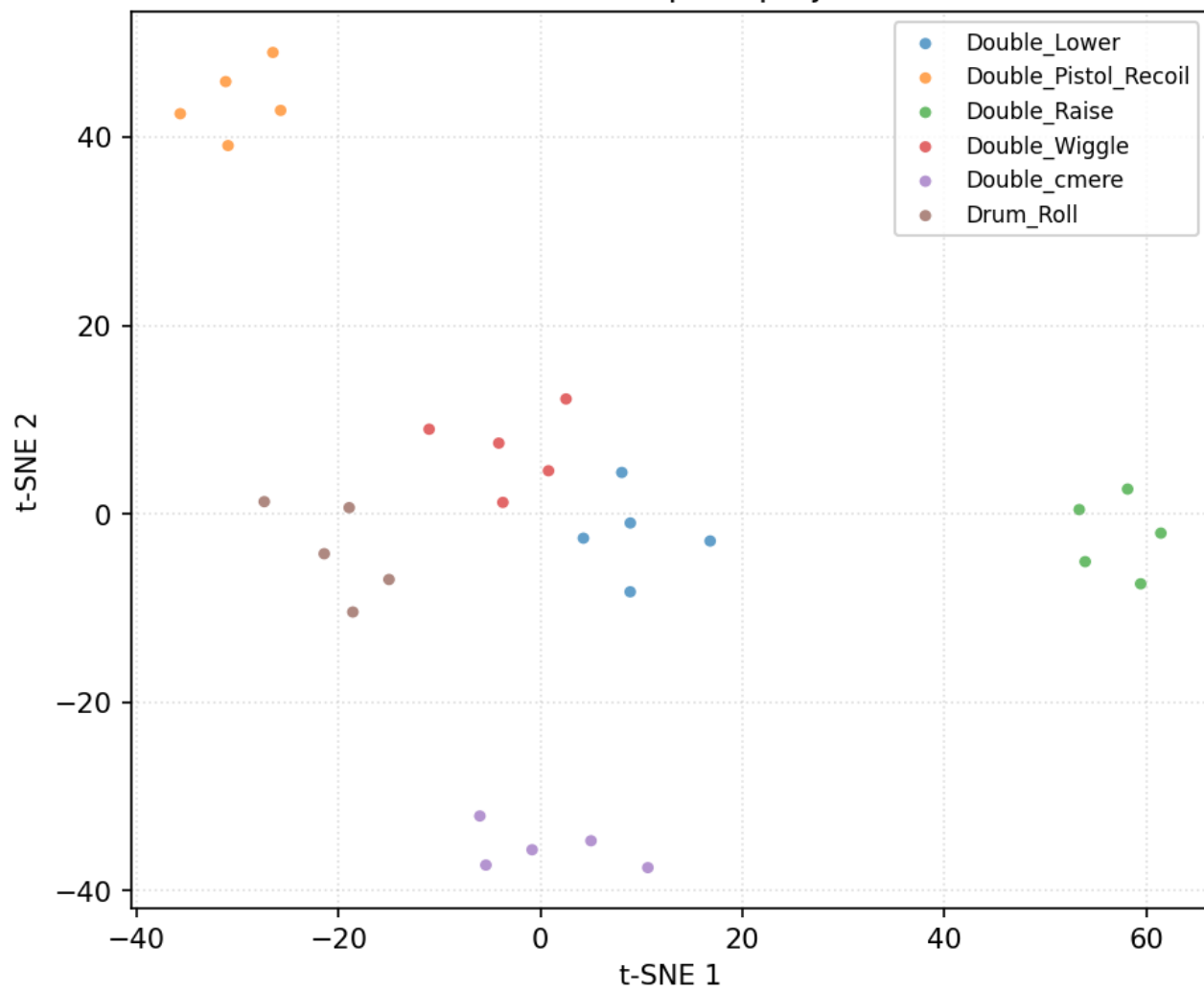


Data diagnostics — user 17_Raquel

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

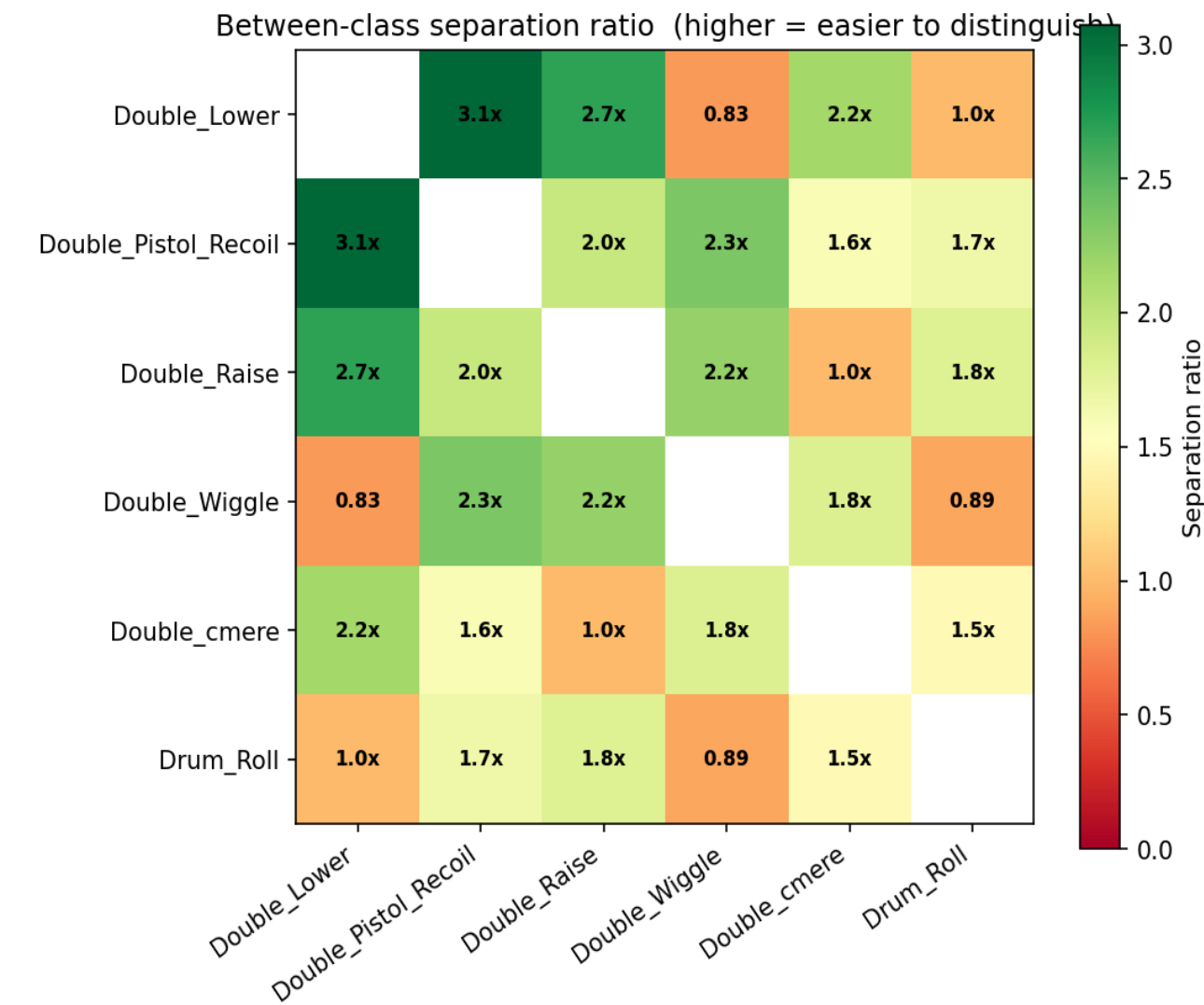
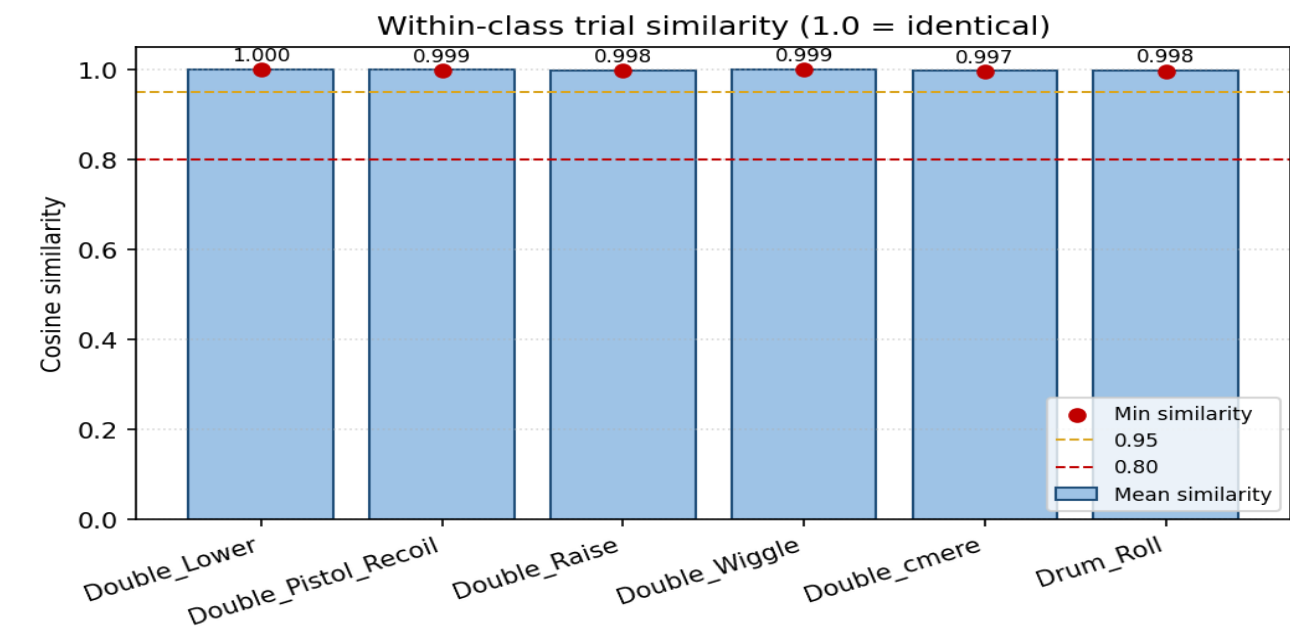


t-SNE feature space projection

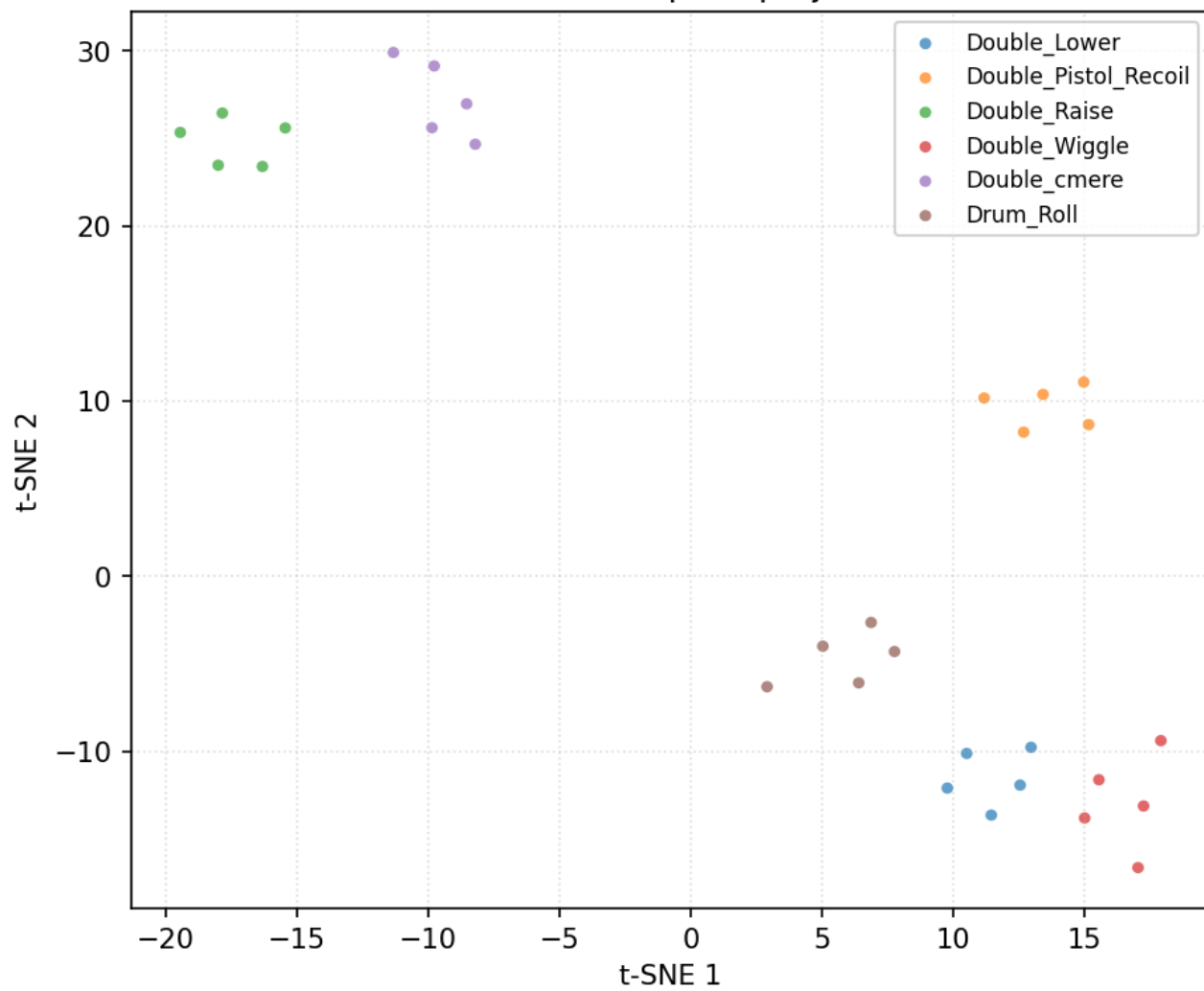


Data diagnostics — user 18_Bridgette

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

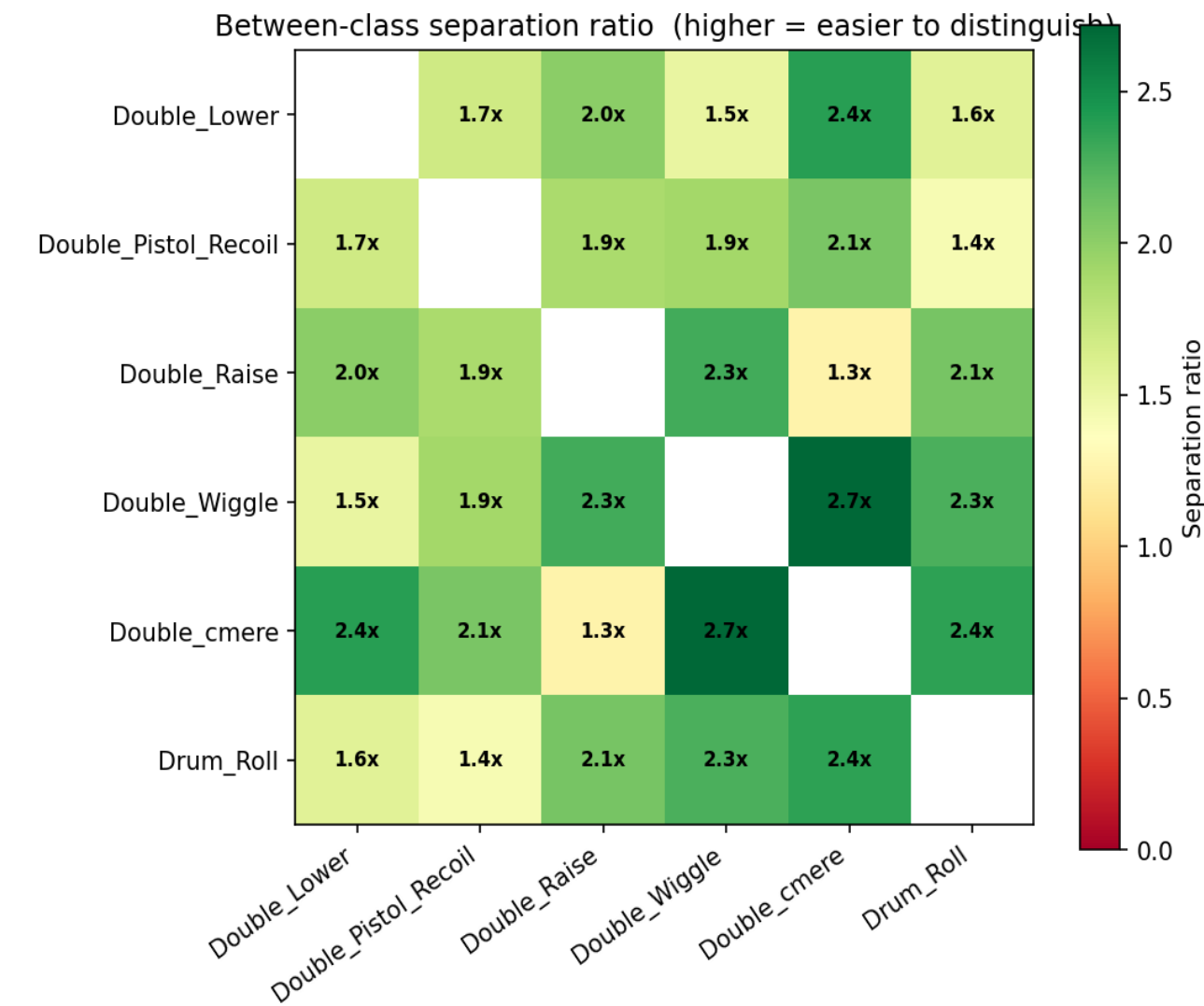
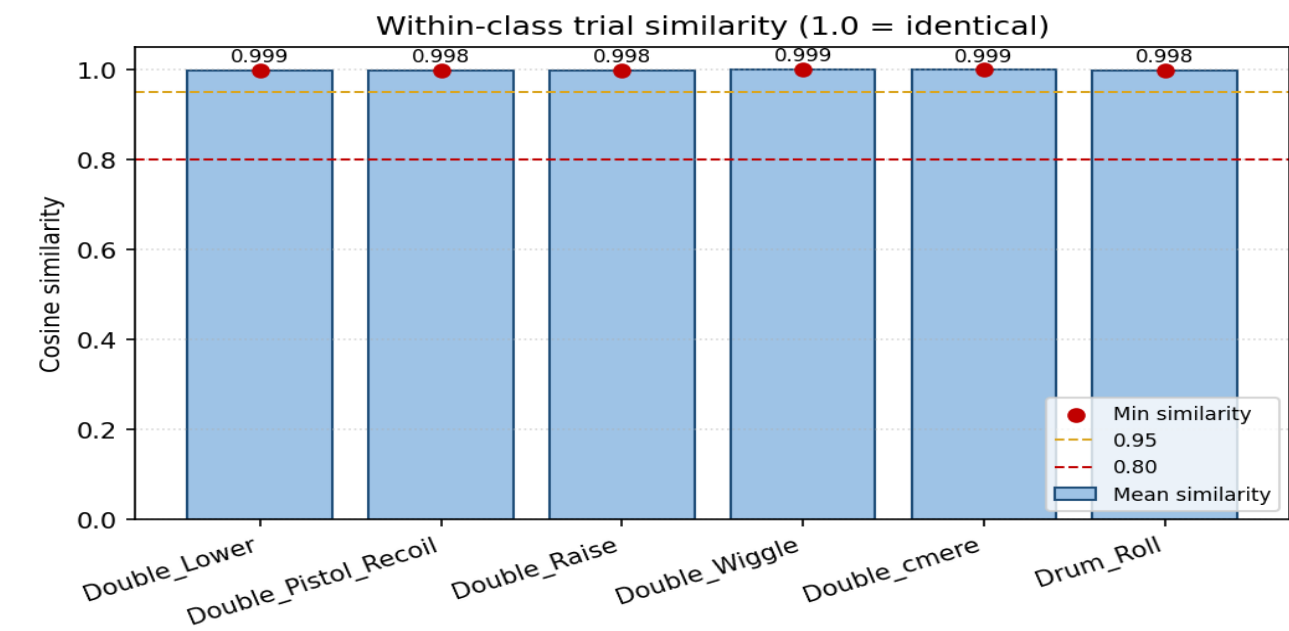


t-SNE feature space projection

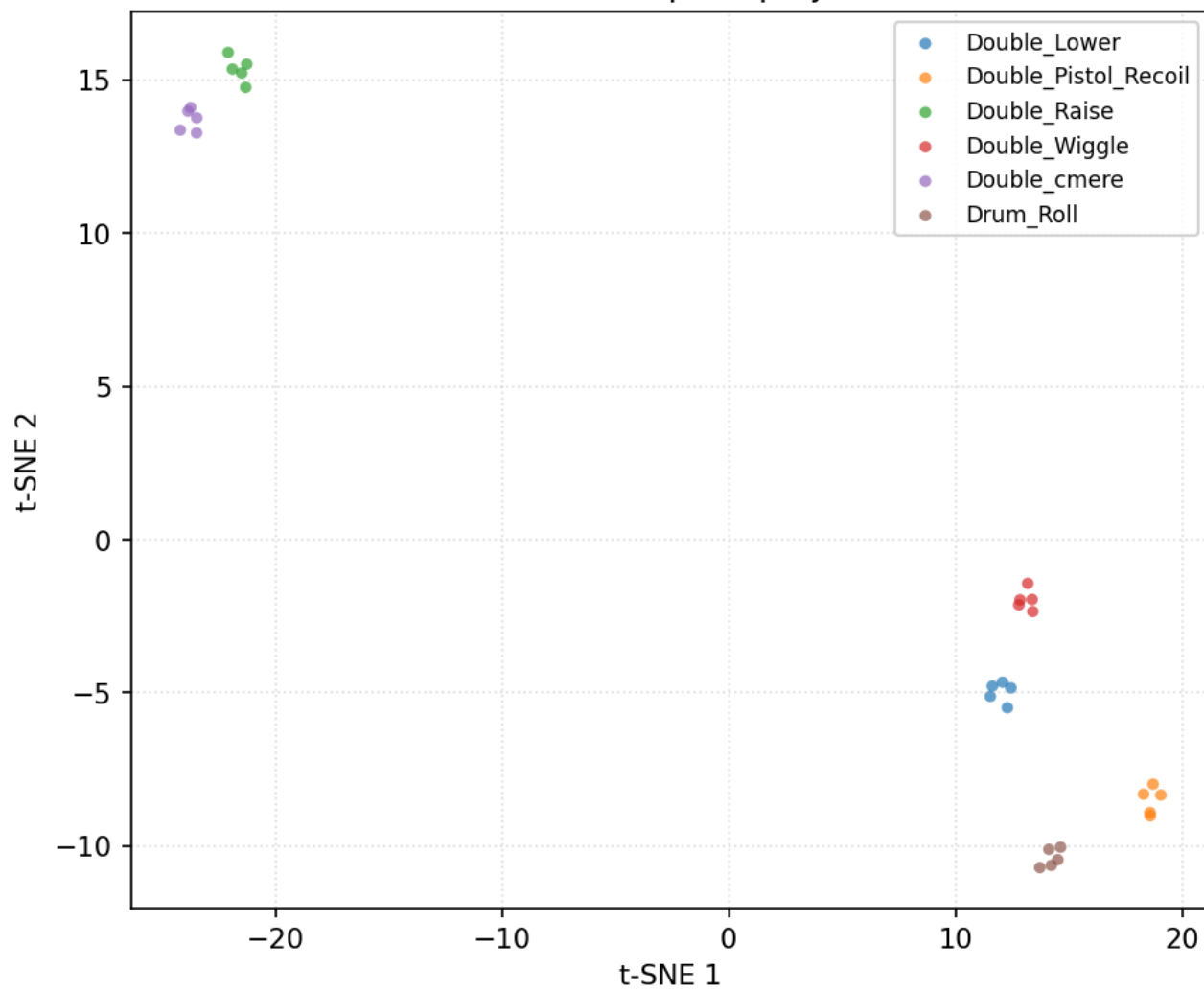


Data diagnostics — user 1_Alán

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

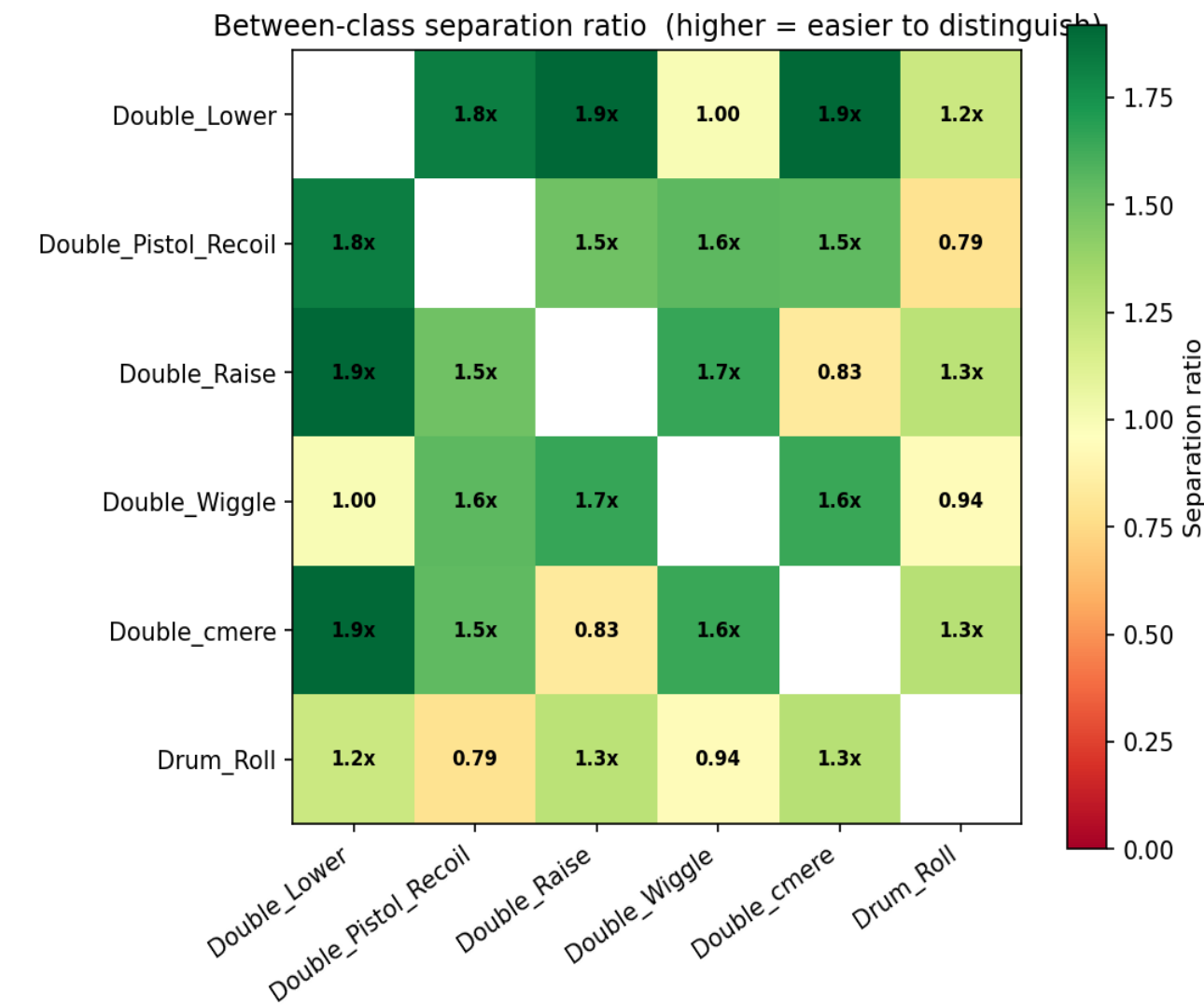
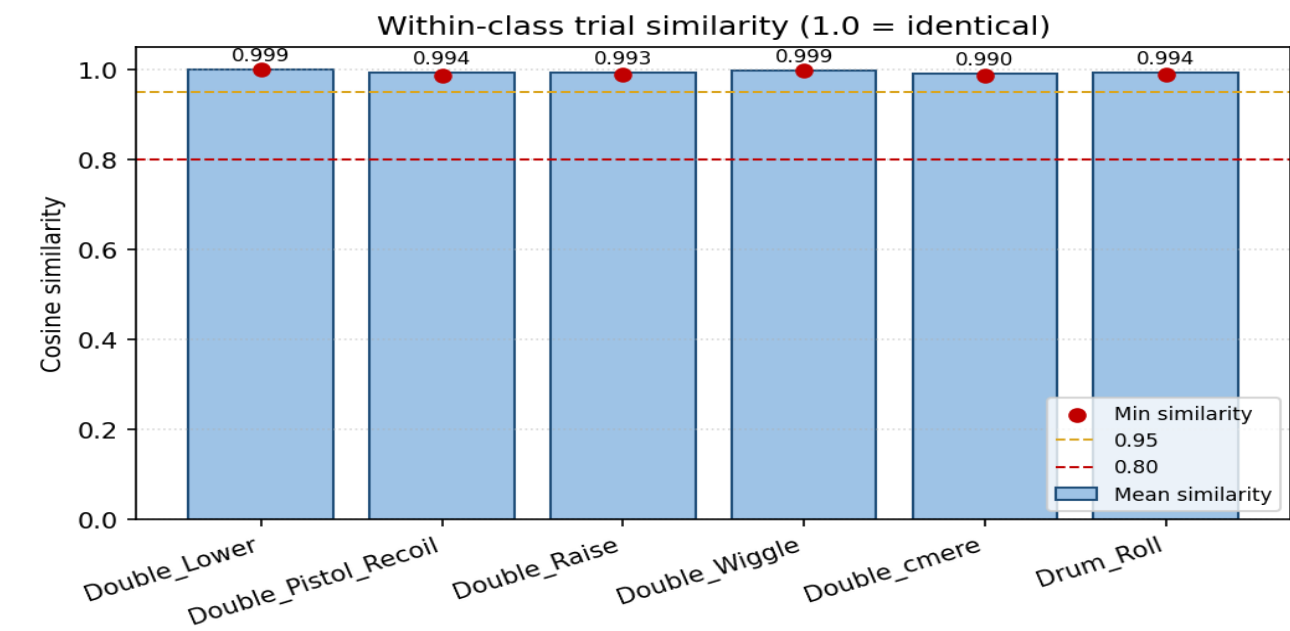


t-SNE feature space projection

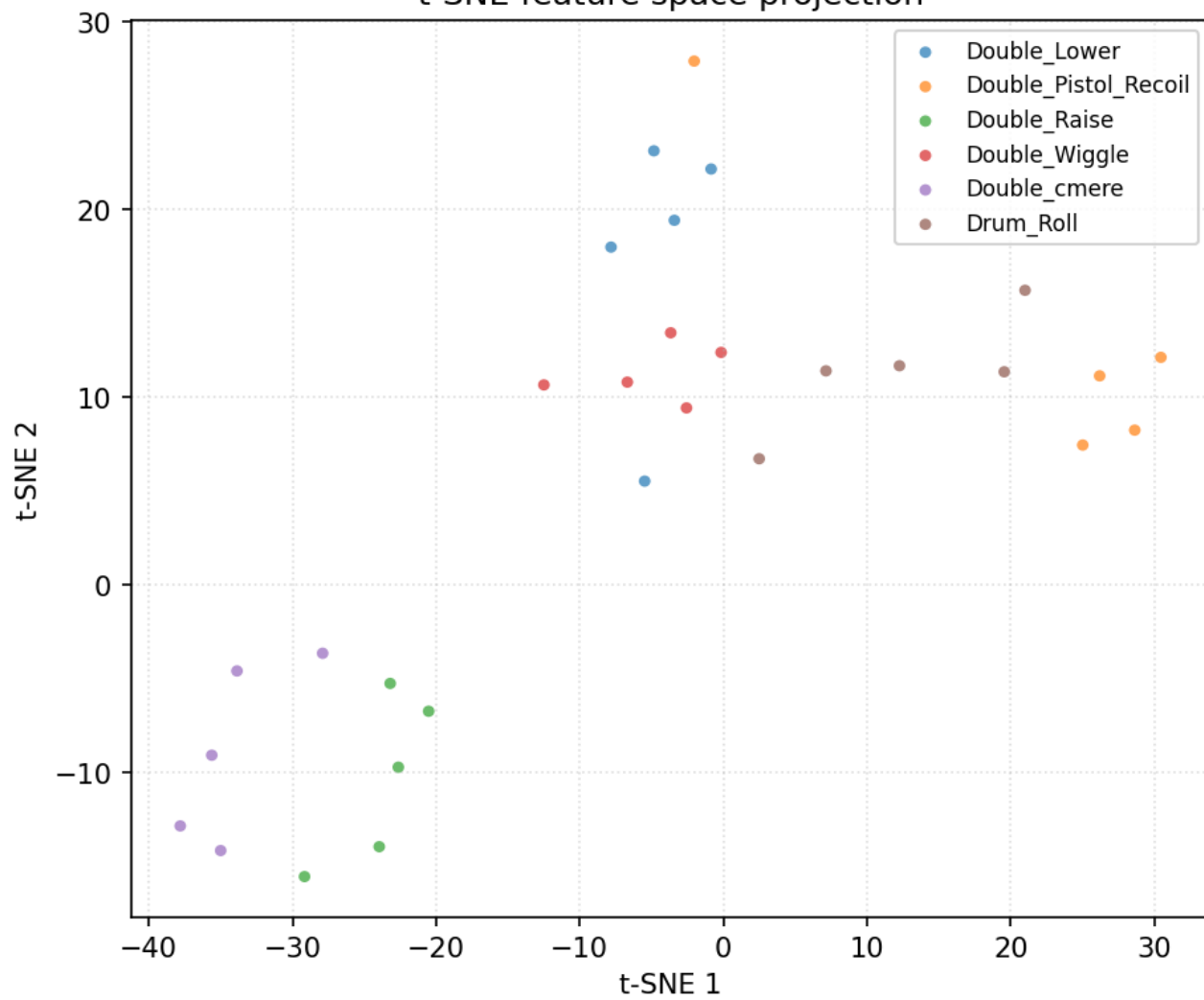


Data diagnostics — user 2_Alex

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

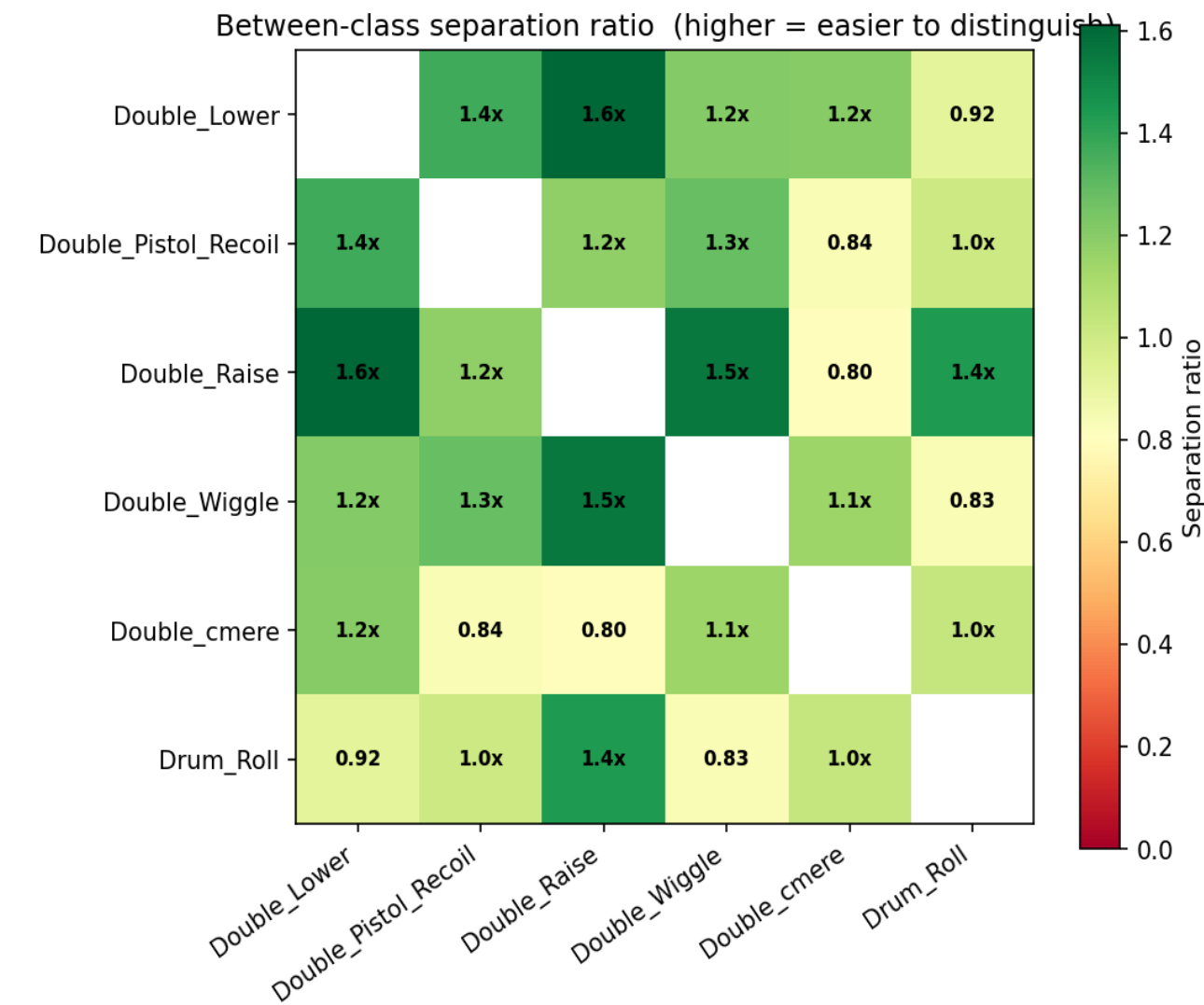
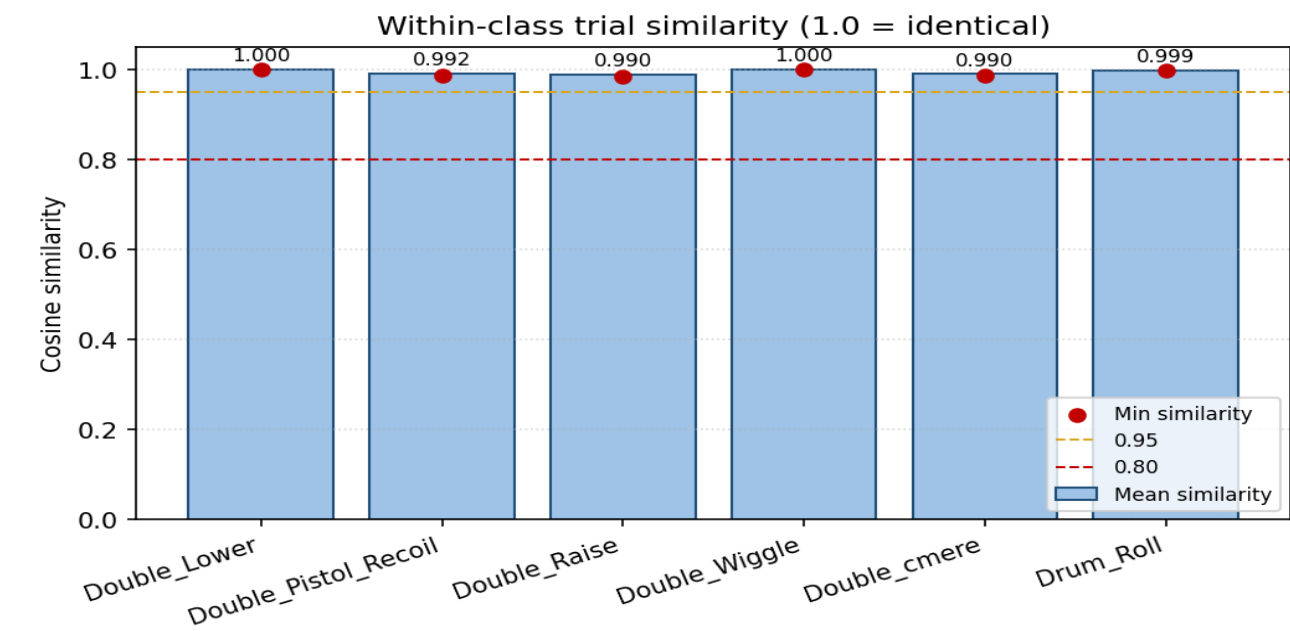


t-SNE feature space projection

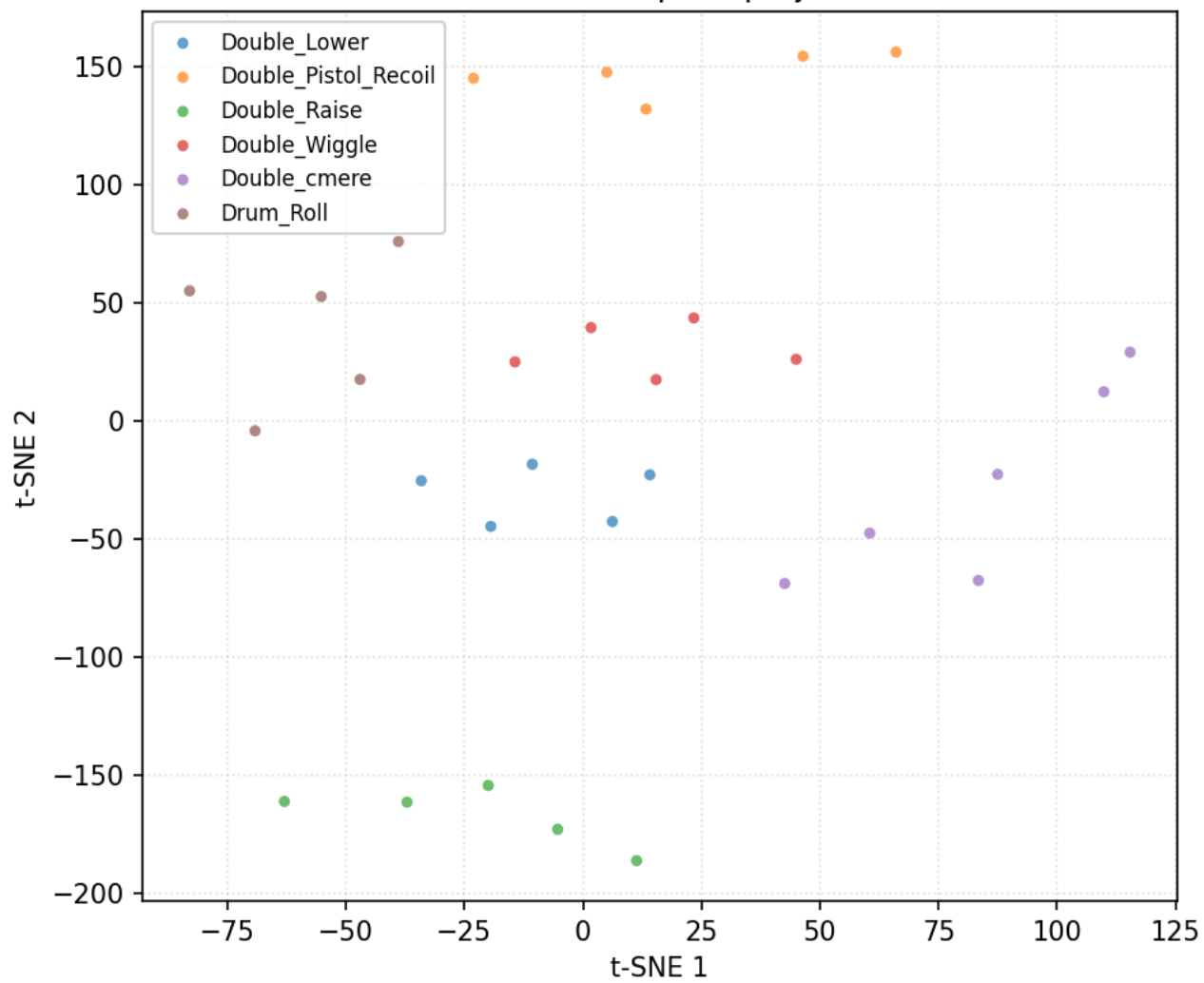


Data diagnostics — user 3_Anghad

31 trials, 6 classes (filtered + resampled, before per-trial normalisation).

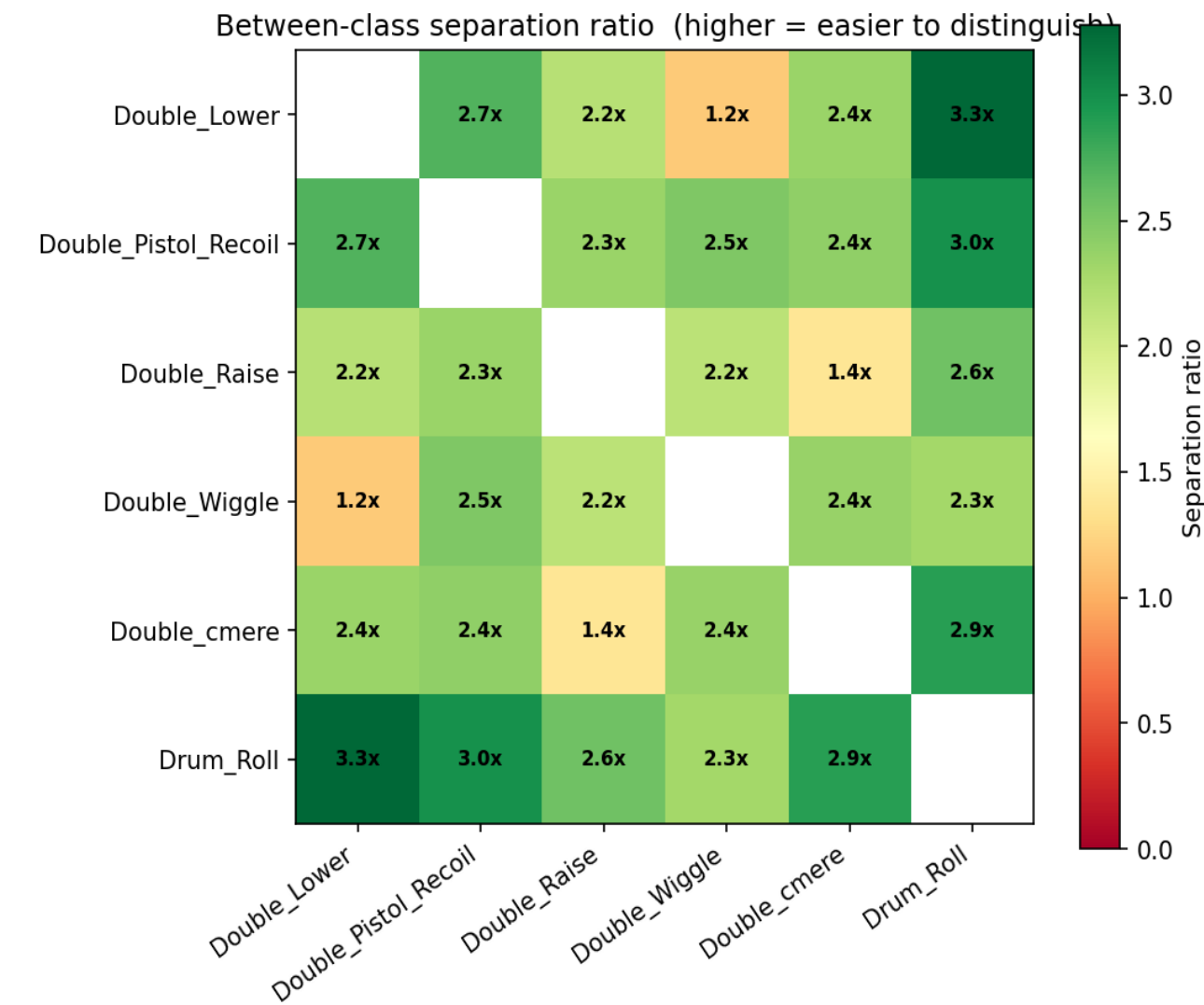
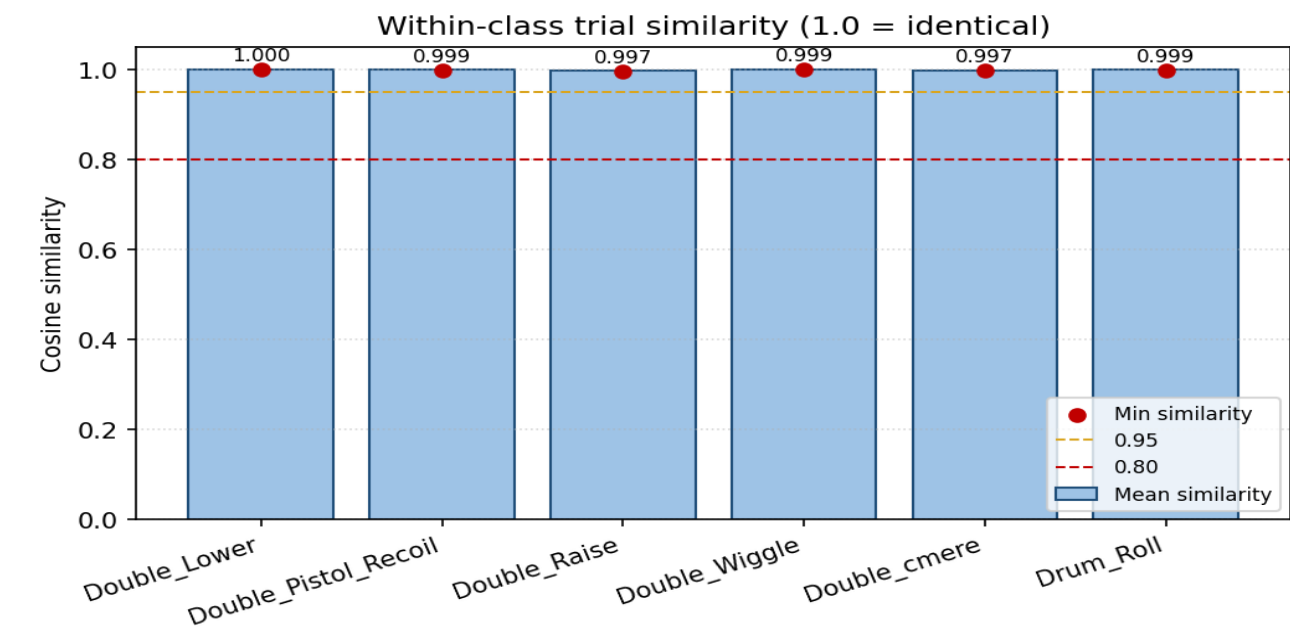


t-SNE feature space projection

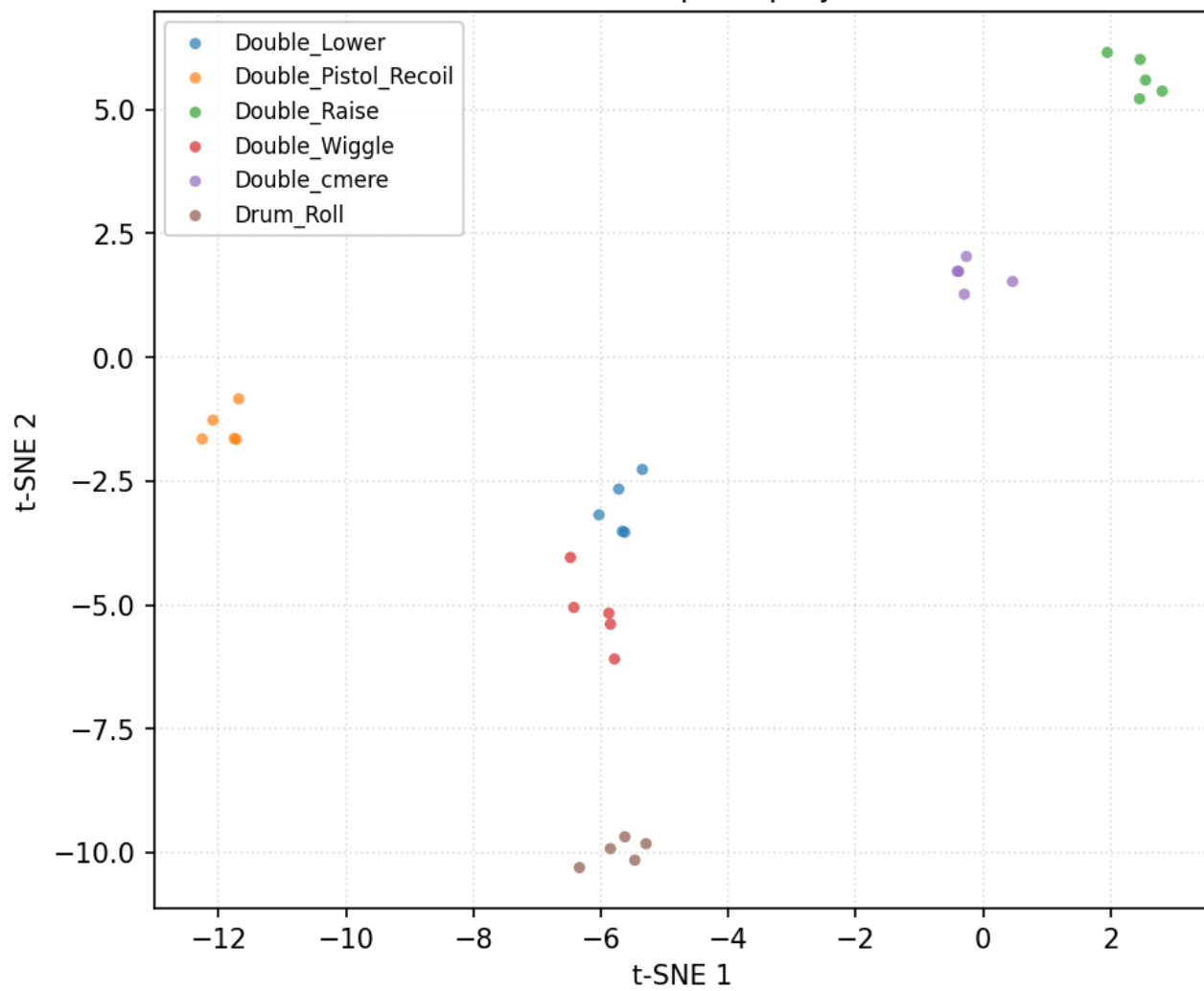


Data diagnostics — user 4_Daniel

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

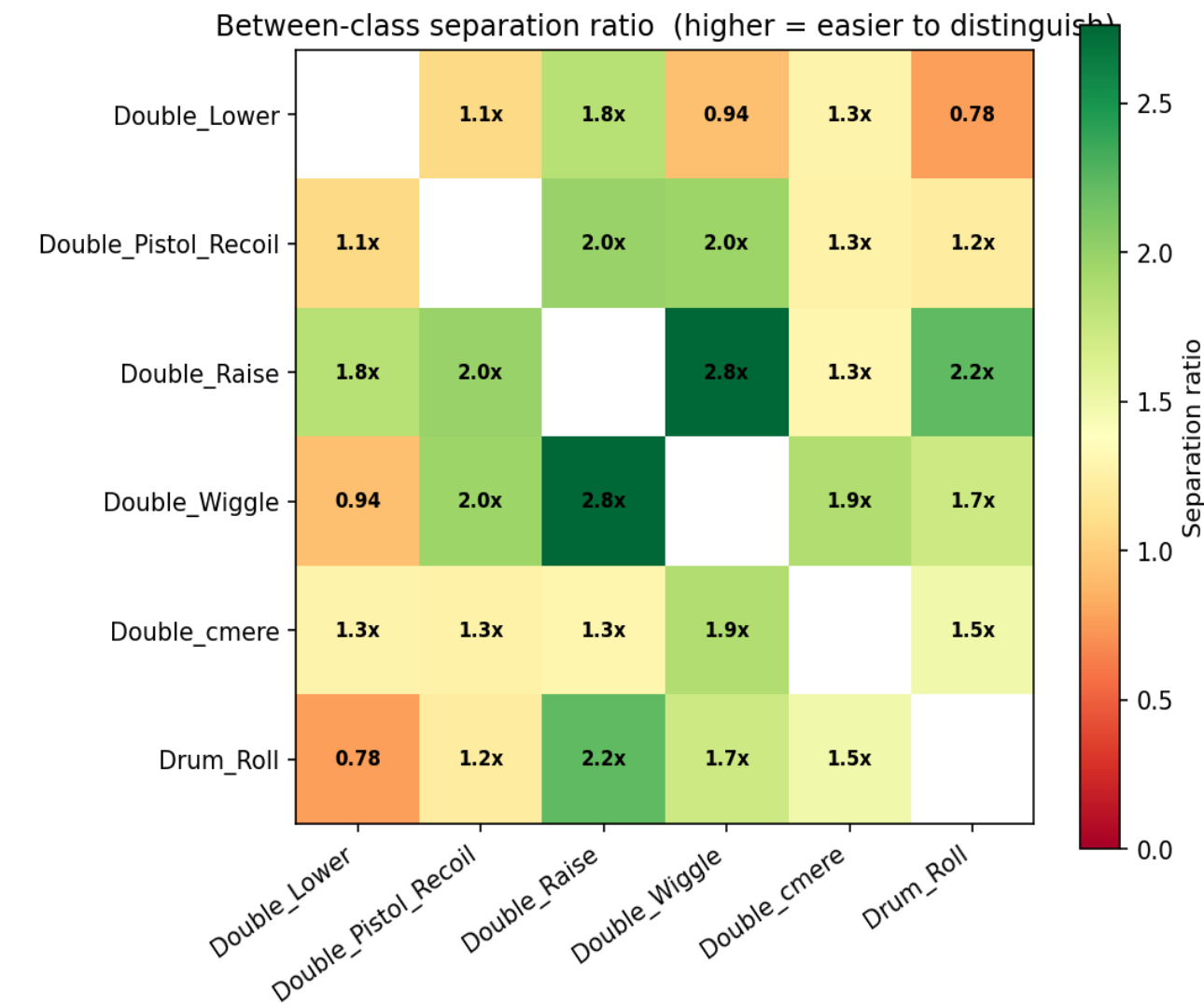
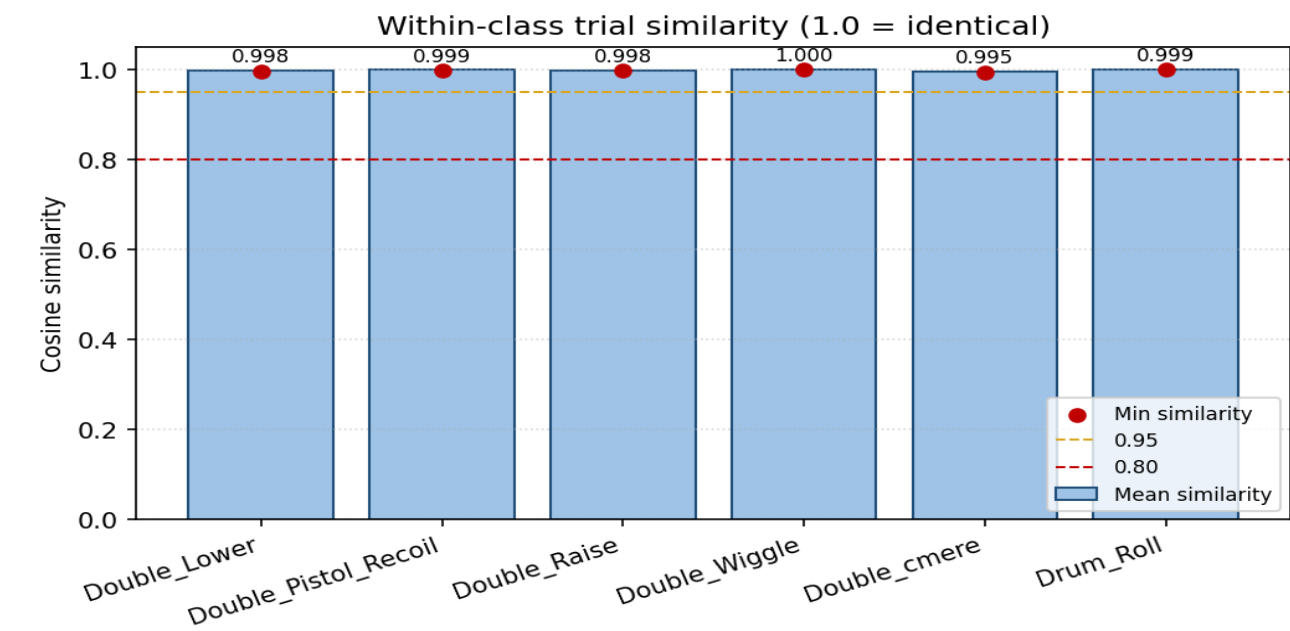


t-SNE feature space projection

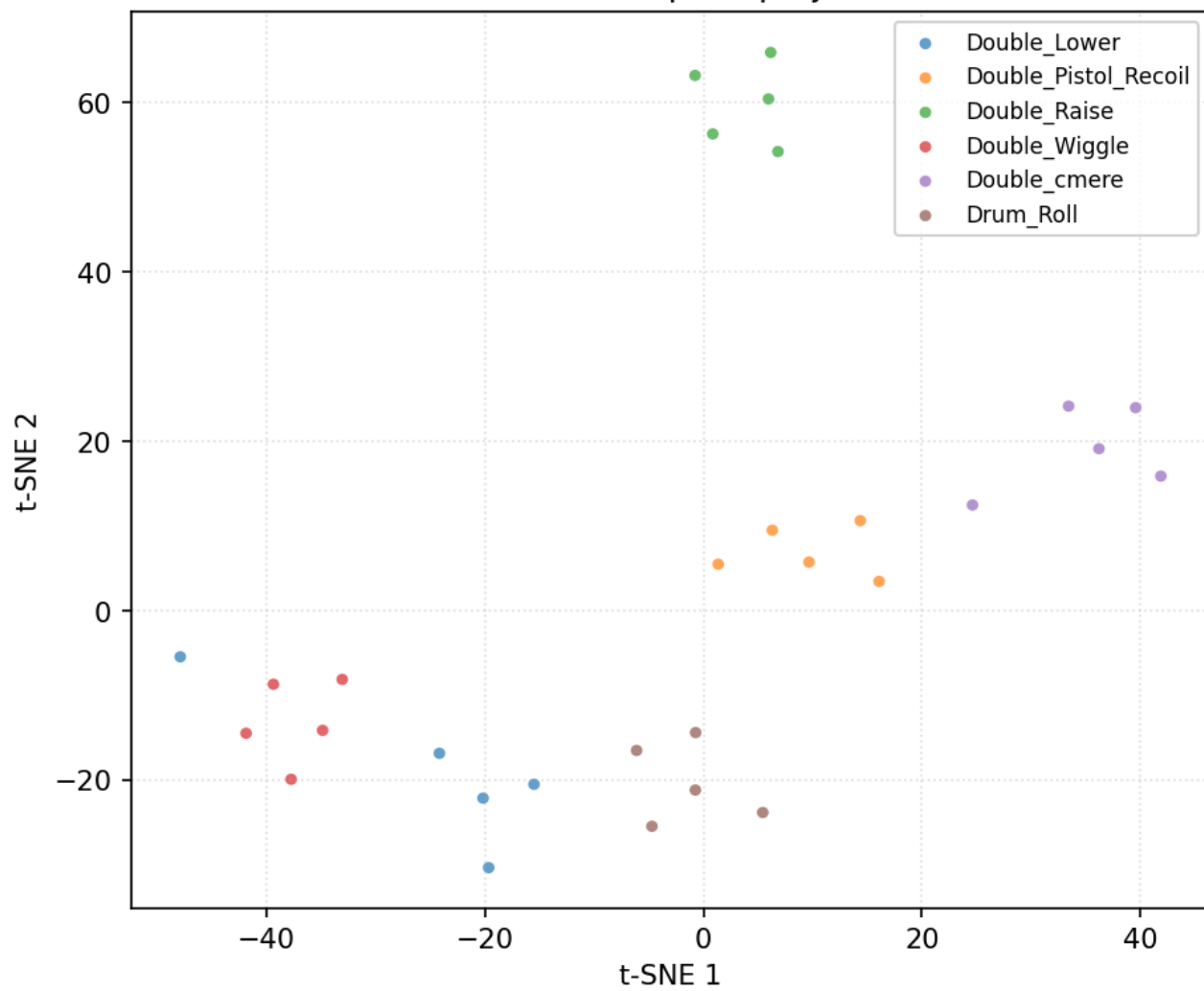


Data diagnostics — user 6_Harry

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

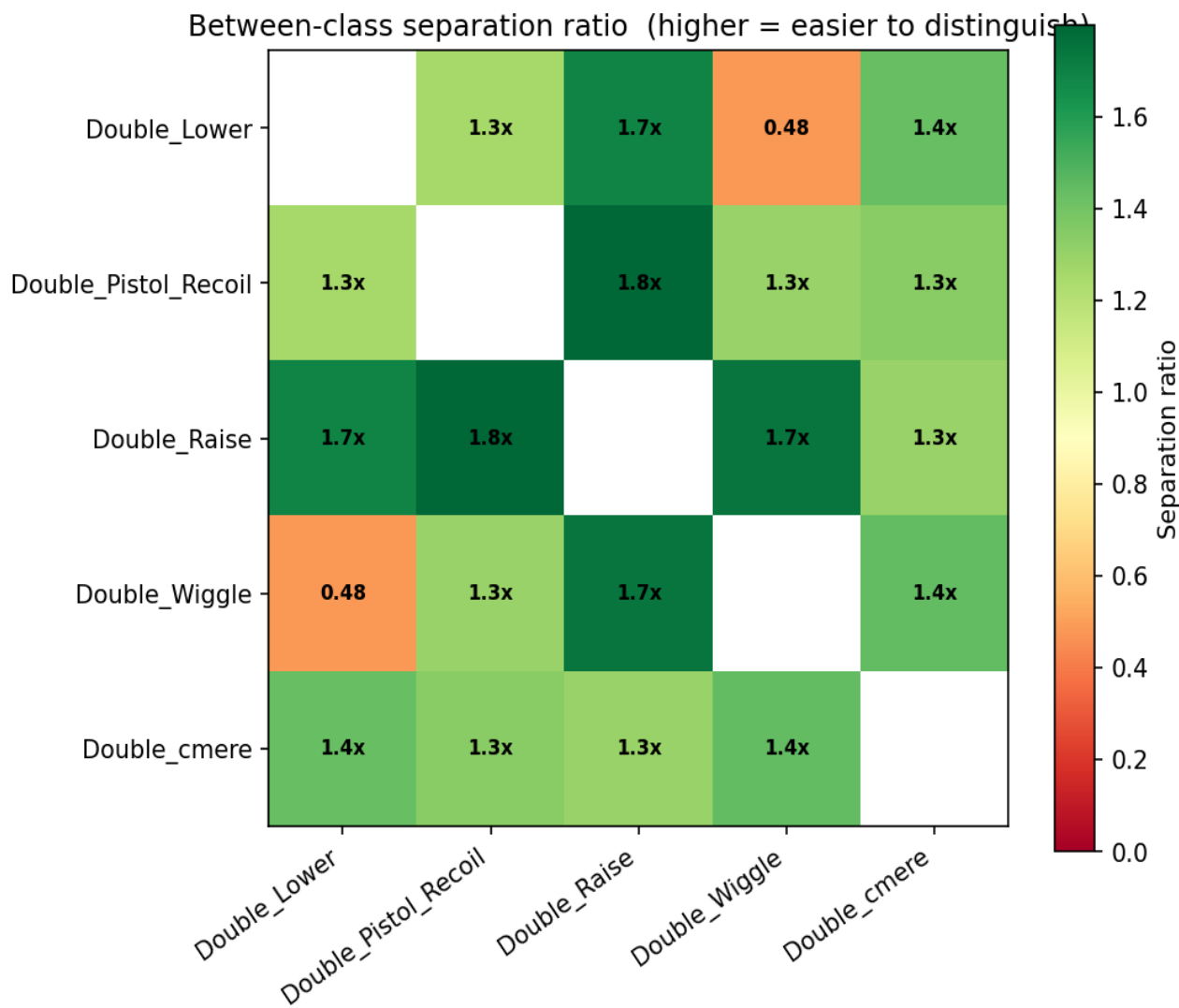
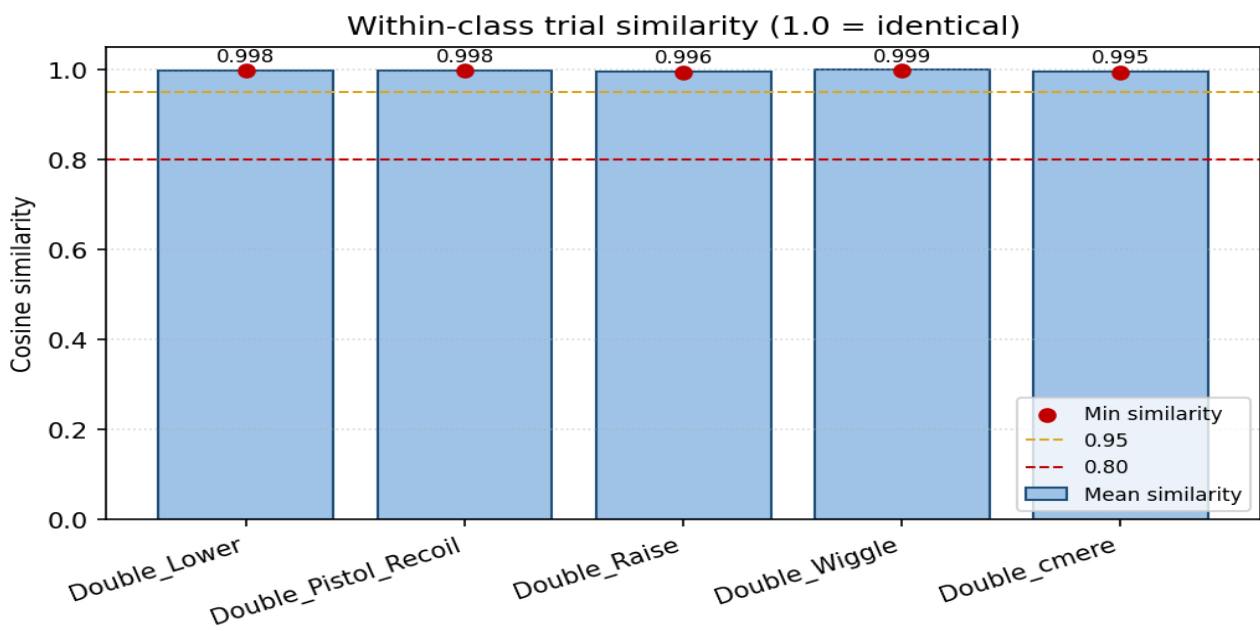


t-SNE feature space projection

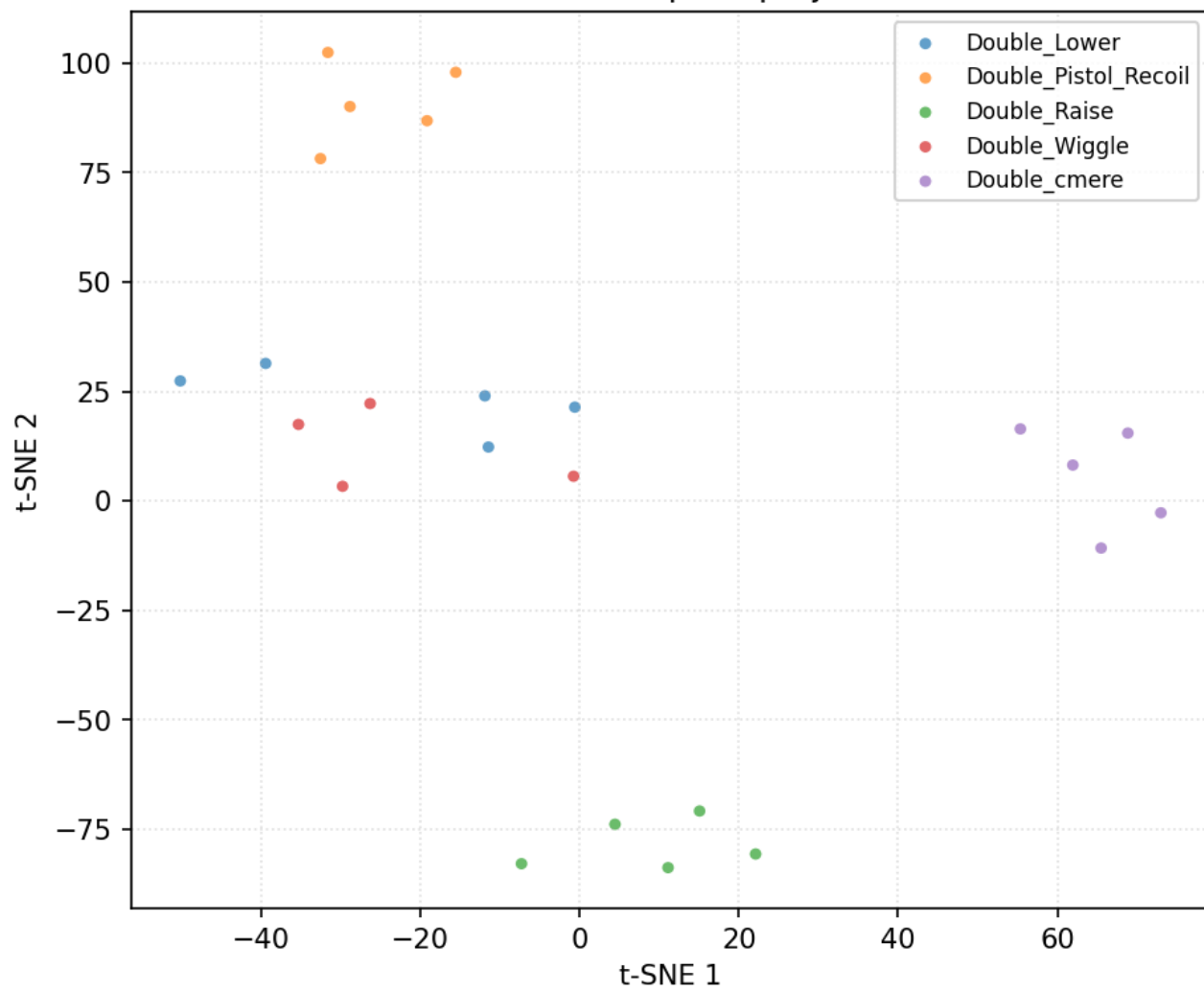


Data diagnostics — user 8_Jestin

24 trials, 5 classes (filtered + resampled, before per-trial normalisation).



t-SNE feature space projection



Variant: BN_GAP

Conv+BN+ReLU ×3 with GlobalAveragePooling1D head — fewer params, less overfit

Trainable params	57,190
LOUO mean ± std	0.9033 ± 0.1309
Folds run	13
Inner-val mode	user
Per-trial norm	none

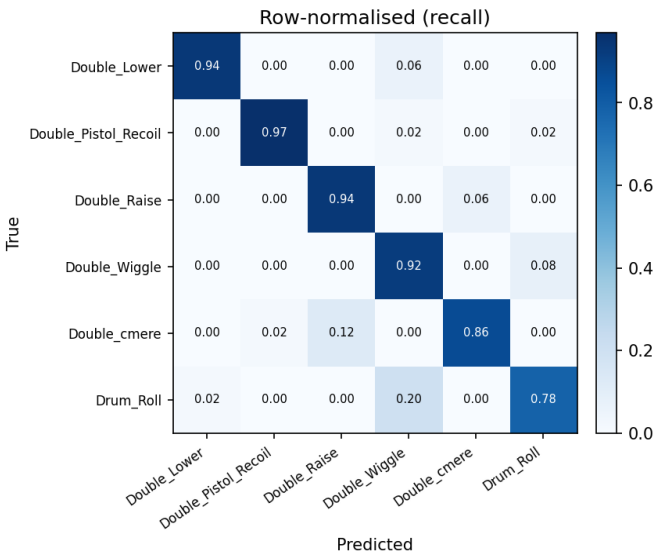
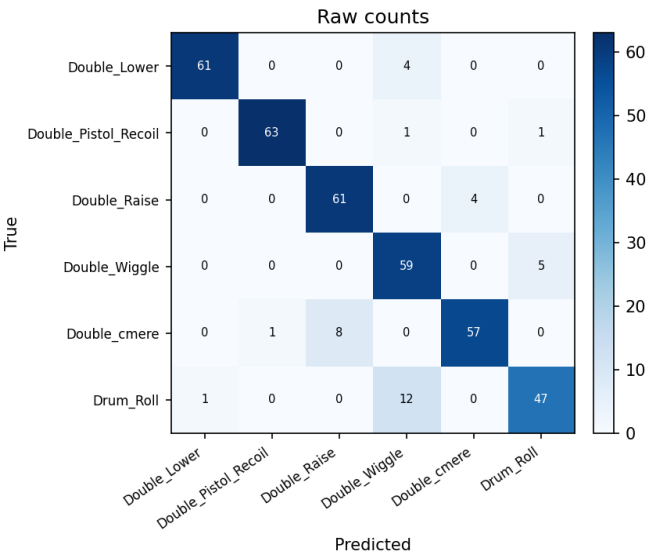
LOUO per-fold results

Fold	Held-out user	Inner-val user	n_train	n_test	Accuracy
1	11_Mansh	12_Marcus	975	30	0.9667
2	12_Marcus	14_StephenV2	975	30	0.8667
3	14_StephenV2	15_Tash	975	30	1.0000
4	15_Tash	16_Bella	975	30	0.7333
5	16_Bella	17_Raquel	975	30	1.0000
6	17_Raquel	18_Bridgette	975	30	1.0000
7	18_Bridgette	1_Alan	975	30	0.8667
8	1_Alan	2_Alex	975	30	0.5333
9	2_Alex	3_Anghad	972	30	0.9667
10	3_Anghad	4_Daniel	972	31	0.9677
11	4_Daniel	6_Harry	975	30	1.0000
12	6_Harry	8_Jestin	993	30	0.9667
13	8_Jestin	11_Mansh	993	24	0.8750

Per-class metrics (aggregated across folds)

Class	Precision	Recall	F1	Support
Double_Lower	0.984	0.938	0.961	65
Double_Pistol_Recoil	0.984	0.969	0.977	65
Double_Raise	0.884	0.938	0.910	65
Double_Wiggle	0.776	0.922	0.843	64
Double_cmere	0.934	0.864	0.898	66
Drum_Roll	0.887	0.783	0.832	60
macro avg	0.908	0.902	0.903	385
weighted avg	0.909	0.904	0.904	385

Confusion matrix (aggregated across folds)



Variant: Deep

Three Conv blocks: Conv1D(32,4)→Conv1D(64,4)→Conv1D(128,3)→Flatten→Dense(128)→Dropout(0.4)→Softmax

Trainable params	203,878
LOUO mean ± std	0.8904 ± 0.1178
Folds run	13
Inner-val mode	user
Per-trial norm	none

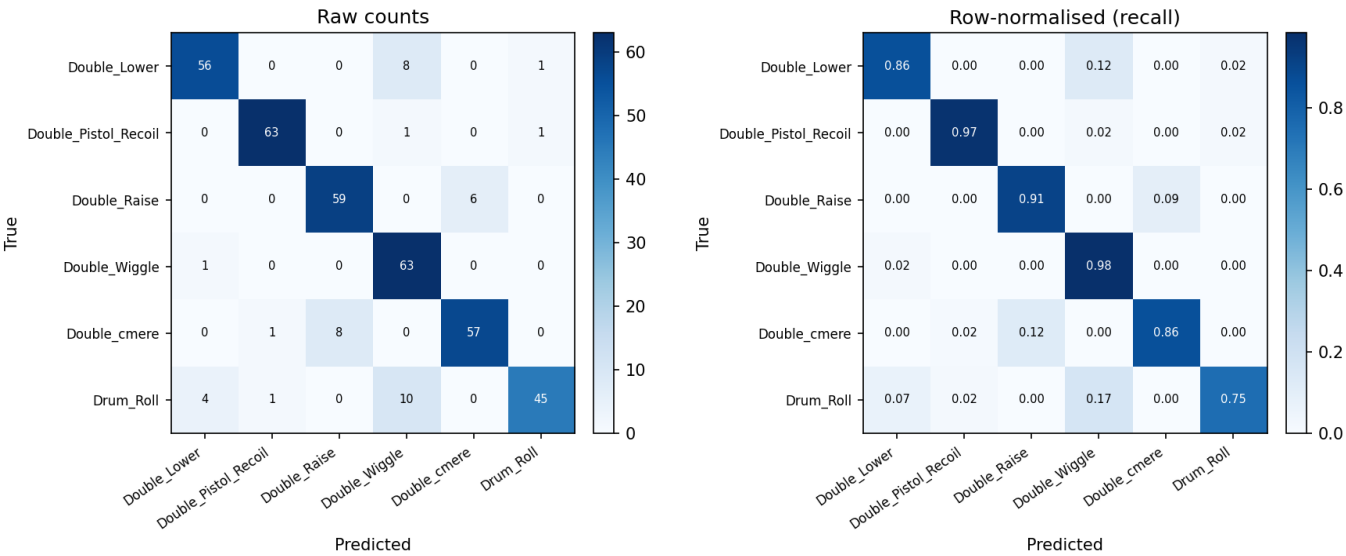
LOUO per-fold results

Fold	Held-out user	Inner-val user	n_train	n_test	Accuracy
1	11_Mansh	12_Marcus	975	30	0.9333
2	12_Marcus	14_StephenV2	975	30	0.8667
3	14_StephenV2	15_Tash	975	30	0.9667
4	15_Tash	16_Bella	975	30	0.6667
5	16_Bella	17_Raquel	975	30	1.0000
6	17_Raquel	18_Bridgette	975	30	1.0000
7	18_Bridgette	1_Alan	975	30	0.7000
8	1_Alan	2_Alex	975	30	0.7000
9	2_Alex	3_Anghad	972	30	0.9667
10	3_Anghad	4_Daniel	972	31	1.0000
11	4_Daniel	6_Harry	975	30	0.9333
12	6_Harry	8_Jestin	993	30	0.9667
13	8_Jestin	11_Mansh	993	24	0.8750

Per-class metrics (aggregated across folds)

Class	Precision	Recall	F1	Support
Double_Lower	0.918	0.862	0.889	65
Double_Pistol_Recoil	0.969	0.969	0.969	65
Double_Raise	0.881	0.908	0.894	65
Double_Wiggle	0.768	0.984	0.863	64
Double_cmere	0.905	0.864	0.884	66
Drum_Roll	0.957	0.750	0.841	60
macro avg	0.900	0.889	0.890	385
weighted avg	0.899	0.891	0.891	385

Confusion matrix (aggregated across folds)



Variant: Baseline

Conv1D(32,k=4)→Pool→Conv1D(64,k=4)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	113,190
LOUO mean ± std	0.8186 ± 0.1802
Folds run	13
Inner-val mode	user
Per-trial norm	none

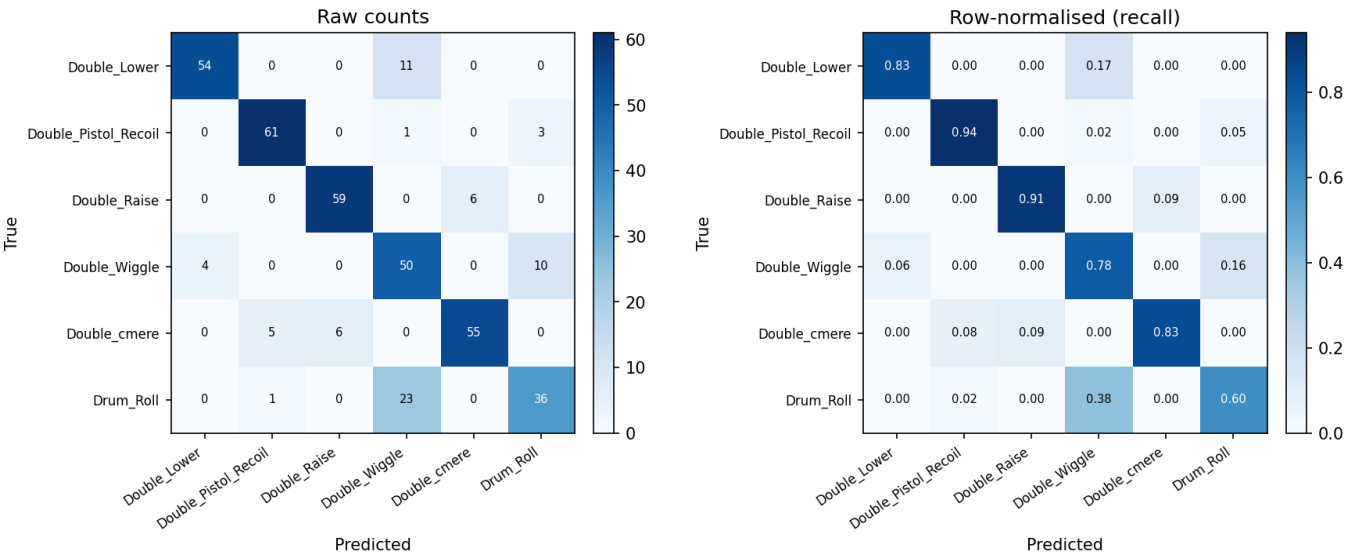
LOUO per-fold results

Fold	Held-out user	Inner-val user	n_train	n_test	Accuracy
1	11_Mansh	12_Marcus	975	30	0.9000
2	12_Marcus	14_StephenV2	975	30	0.8667
3	14_StephenV2	15_Tash	975	30	0.8333
4	15_Tash	16_Bella	975	30	0.8333
5	16_Bella	17_Raquel	975	30	0.8333
6	17_Raquel	18_Bridgette	975	30	1.0000
7	18_Bridgette	1_Alan	975	30	0.6667
8	1_Alan	2_Alex	975	30	0.5000
9	2_Alex	3_Anghad	972	30	0.9667
10	3_Anghad	4_Daniel	972	31	1.0000
11	4_Daniel	6_Harry	975	30	0.9667
12	6_Harry	8_Jestin	993	30	0.4000
13	8_Jestin	11_Mansh	993	24	0.8750

Per-class metrics (aggregated across folds)

Class	Precision	Recall	F1	Support
Double_Lower	0.931	0.831	0.878	65
Double_Pistol_Recoil	0.910	0.938	0.924	65
Double_Raise	0.908	0.908	0.908	65
Double_Wiggle	0.588	0.781	0.671	64
Double_cmere	0.902	0.833	0.866	66
Drum_Roll	0.735	0.600	0.661	60
macro avg	0.829	0.815	0.818	385
weighted avg	0.831	0.818	0.821	385

Confusion matrix (aggregated across folds)



Variant: WideKernel

Conv1D(32,k=8)→Pool→Conv1D(64,k=8)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	131,622
LOUO mean ± std	0.8109 ± 0.2070
Folds run	13
Inner-val mode	user
Per-trial norm	none

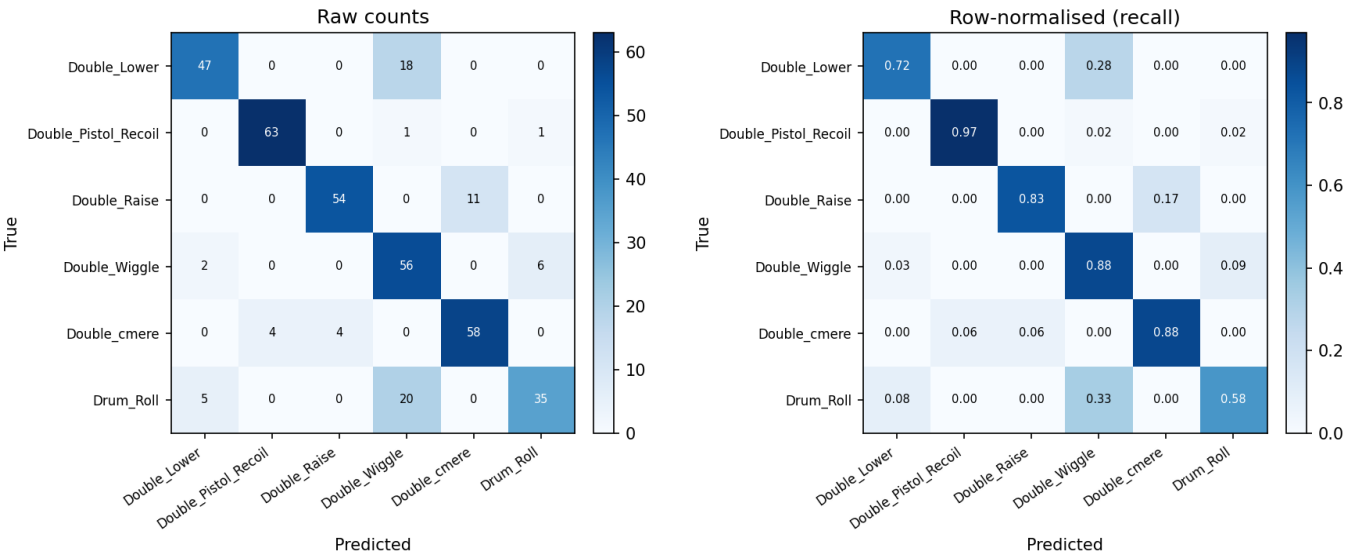
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3	14_StephenV2	15_Tash	975	30	0.8333
4	15_Tash	16_Bella	975	30	0.7667
5	16_Bella	17_Raquel	975	30	0.9667
6	17_Raquel	18_Bridgette	975	30	0.8667
7	18_Bridgette	1_Alan	975	30	0.5000
8	1_Alan	2_Alex	975	30	0.2667
9	2_Alex	3_Anghad	972	30	0.9667
10	3_Anghad	4_Daniel	972	31	1.0000
11	4_Daniel	6_Harry	975	30	1.0000
12	6_Harry	8_Jestin	993	30	0.9667
13	8_Jestin	11_Mansh	993	24	0.7083

Per-class metrics (aggregated across folds)

Class	Precision	Recall	F1	Support
Double_Lower	0.870	0.723	0.790	65
Double_Pistol_Recoil	0.940	0.969	0.955	65
Double_Raise	0.931	0.831	0.878	65
Double_Wiggle	0.589	0.875	0.704	64
Double_cmere	0.841	0.879	0.859	66
Drum_Roll	0.833	0.583	0.686	60
macro avg	0.834	0.810	0.812	385
weighted avg	0.835	0.813	0.814	385

Confusion matrix (aggregated across folds)



Variant: CNN_BiLSTM

Conv1D(32,4)→Pool→Conv1D(64,4)→Pool→Dropout→BiLSTM(64)→Dense(64)→Dropout→Softmax

Trainable params	105,510
LOUO mean ± std	0.8083 ± 0.2229
Folds run	13
Inner-val mode	user
Per-trial norm	none

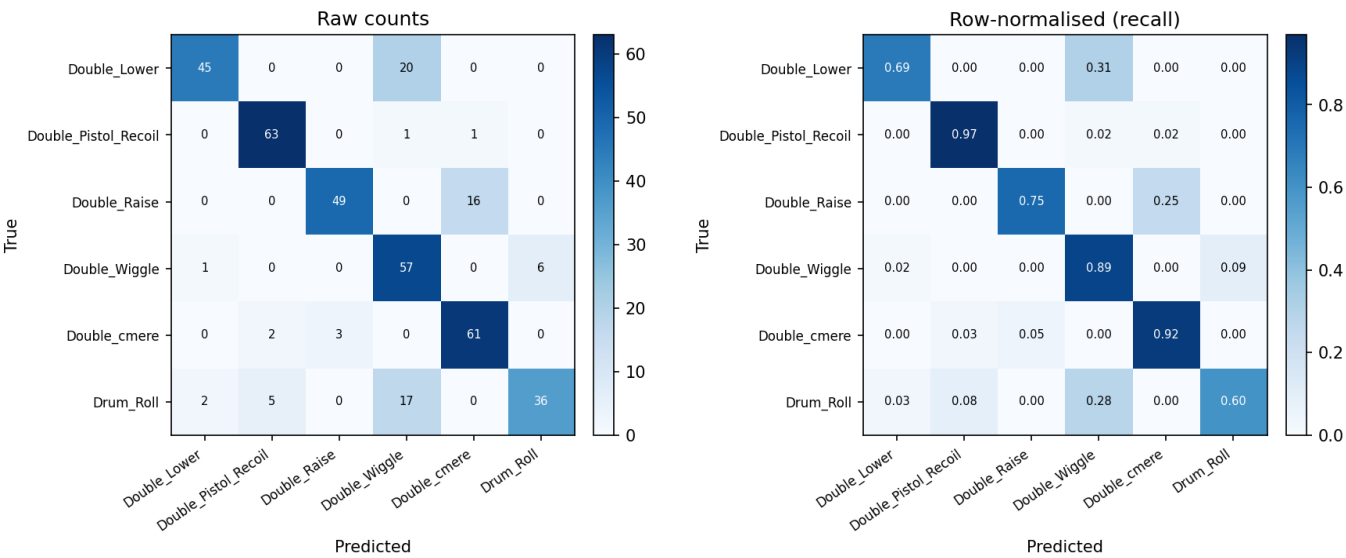
LOUO per-fold results

Fold	Held-out user	Inner-val user	n_train	n_test	Accuracy
1	11_Mansh	12_Marcus	975	30	1.0000
2	12_Marcus	14_StephenV2	975	30	0.9667
3	14_StephenV2	15_Tash	975	30	0.5000
4	15_Tash	16_Bella	975	30	0.6000
5	16_Bella	17_Raquel	975	30	0.9667
6	17_Raquel	18_Bridgette	975	30	0.8000
7	18_Bridgette	1_Alan	975	30	0.5000
8	1_Alan	2_Alex	975	30	0.3667
9	2_Alex	3_Anghad	972	30	0.9333
10	3_Anghad	4_Daniel	972	31	1.0000
11	4_Daniel	6_Harry	975	30	1.0000
12	6_Harry	8_Jestin	993	30	1.0000
13	8_Jestin	11_Mansh	993	24	0.8750

Per-class metrics (aggregated across folds)

Class	Precision	Recall	F1	Support
Double_Lower	0.938	0.692	0.796	65
Double_Pistol_Recoil	0.900	0.969	0.933	65
Double_Raise	0.942	0.754	0.838	65
Double_Wiggle	0.600	0.891	0.717	64
Double_cmere	0.782	0.924	0.847	66
Drum_Roll	0.857	0.600	0.706	60
macro avg	0.837	0.805	0.806	385
weighted avg	0.837	0.808	0.808	385

Confusion matrix (aggregated across folds)



Variant: Shallow

Single Conv block: Conv1D(32,k=5)→Pool→Flatten→Dense(32)→Dropout(0.3)→Softmax

Trainable params	72,454
LOUO mean ± std	0.7692 ± 0.1928
Folds run	13
Inner-val mode	user
Per-trial norm	none

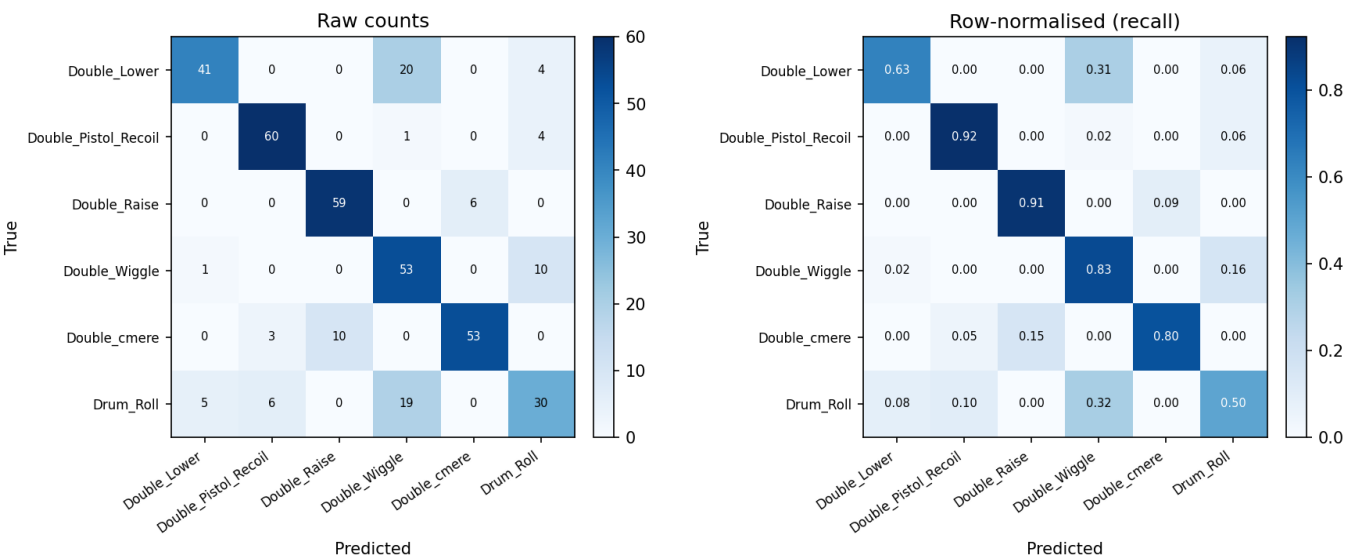
LOUO per-fold results

Fold	Held-out user	Inner-val user	n_train	n_test	Accuracy
1	11_Mansh	12_Marcus	975	30	0.8333
2	12_Marcus	14_StephenV2	975	30	0.8333
3	14_StephenV2	15_Tash	975	30	0.7000
4	15_Tash	16_Bella	975	30	0.5667
5	16_Bella	17_Raquel	975	30	0.8667
6	17_Raquel	18_Bridgette	975	30	0.9667
7	18_Bridgette	1_Alan	975	30	0.6333
8	1_Alan	2_Alex	975	30	0.3667
9	2_Alex	3_Anghad	972	30	0.9000
10	3_Anghad	4_Daniel	972	31	1.0000
11	4_Daniel	6_Harry	975	30	1.0000
12	6_Harry	8_Jestin	993	30	0.5000
13	8_Jestin	11_Mansh	993	24	0.8333

Per-class metrics (aggregated across folds)

Class	Precision	Recall	F1	Support
Double_Lower	0.872	0.631	0.732	65
Double_Pistol_Recoil	0.870	0.923	0.896	65
Double_Raise	0.855	0.908	0.881	65
Double_Wiggle	0.570	0.828	0.675	64
Double_cmere	0.898	0.803	0.848	66
Drum_Roll	0.625	0.500	0.556	60
macro avg	0.782	0.765	0.764	385
weighted avg	0.785	0.769	0.768	385

Confusion matrix (aggregated across folds)



Variant: CNN_LSTM

Conv1D(32,4)→Pool→Conv1D(64,4)→Pool→Dropout→LSTM(64)→Dense(64)→Dropout→Softmax

Trainable params	68,390
LOUO mean ± std	0.7628 ± 0.1861
Folds run	13
Inner-val mode	user
Per-trial norm	none

LOUO per-fold results

Fold	Held-out user	Inner-val user	n_train	n_test	Accuracy
1	11_Mansh	12_Marcus	975	30	0.7333
2	12_Marcus	14_StephenV2	975	30	0.8667
3	14_StephenV2	15_Tash	975	30	0.7333
4	15_Tash	16_Bella	975	30	0.7667
5	16_Bella	17_Raquel	975	30	0.8333
6	17_Raquel	18_Bridgette	975	30	0.8333
7	18_Bridgette	1_Alan	975	30	0.7667
8	1_Alan	2_Alex	975	30	0.3667
9	2_Alex	3_Anghad	972	30	0.9333
10	3_Anghad	4_Daniel	972	31	1.0000
11	4_Daniel	6_Harry	975	30	1.0000
12	6_Harry	8_Jestin	993	30	0.6667
13	8_Jestin	11_Mansh	993	24	0.4167

Per-class metrics (aggregated across folds)

Class	Precision	Recall	F1	Support
Double_Lower	1.000	0.646	0.785	65
Double_Pistol_Recoil	0.838	0.877	0.857	65
Double_Raise	0.884	0.938	0.910	65
Double_Wiggle	0.528	0.734	0.614	64
Double_cmere	0.909	0.758	0.826	66
Drum_Roll	0.629	0.650	0.639	60
macro avg	0.798	0.767	0.772	385
weighted avg	0.801	0.769	0.774	385

Confusion matrix (aggregated across folds)

